

Bootstrap based inference for sparse high-dimensional time series models

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Fitting sparse models to high-dimensional time series is an important area of statistical inference. In this paper, we consider sparse vector autoregressive models and develop appropriate bootstrap methods to infer properties of such processes. Our bootstrap methodology generates pseudo time series using a model-based bootstrap procedure which involves an estimated, sparsified version of the underlying vector autoregressive model. Inference is performed using so-called de-sparsified or de-biased estimators of the autoregressive model parameters. We derive the asymptotic distribution of such estimators in the time series context and establish asymptotic validity of the bootstrap procedure proposed for estimation and, appropriately modified, for testing purposes. In particular, we focus on testing that large groups of autoregressive coefficients equal zero. Our theoretical results are complemented by simulations which investigate the finite sample performance of the bootstrap methodology proposed. A real-life data application is also presented.

Keywords: De-sparsified estimators; testing; vector autoregressive models

1. Introduction

Statistical analysis of high-dimensional time series has attracted considerable interest during the last decades. Initiated by developments in the i.i.d., mainly regression, set-up, statistical methods have been proposed to select and to estimate non-zero parameters in the context of sparse high-dimensional time series models by means of regularized-type estimators. To be more specific, consider a p dimensional centered stochastic process $\{X_t, t \in \mathbb{Z}\}$, where the random vector X_t is generated via a d th order vector autoregressive (VAR(d)) model,

$$X_t = \sum_{s=1}^d A^{(s)} X_{t-s} + \varepsilon_t. \quad (1)$$

Here $A^{(s)}$, $s = 1, 2, \dots, d$, are $p \times p$ coefficient matrices while the ε_t 's are assumed to be independent and identically distributed (i.i.d.) innovations with $E(\varepsilon_t) = 0$, $\text{Var}(\varepsilon_t) = \Sigma_\varepsilon$, in short, $\varepsilon_t \sim (0, \Sigma_\varepsilon)$. Assume that the process is stationary and causal, that is $\det(\mathcal{A}(z)) \neq 0$ for all $z \leq 1$, where $\mathcal{A}(z) = I - \sum_{s=1}^d A^{(s)} z^s$. Model (1) has dp^2 unknown parameters in the matrices $A^{(s)}$, $s = 1, \dots, d$ and $p(p+1)/2$ unknown parameters in the innovation covariance matrix Σ_ε . Hence, the total number of unknown parameters is $q = p^2(d+1/2) + p/2 = O(dp^2)$. Suppose that a time series $X_1, X_2, \dots, X_n$ stemming from $\{X_t, t \in \mathbb{Z}\}$ has been observed. If the number of parameters is small in the sense that $q \ll n$, then inference for such processes is a well developed and well understood area in multivariate time series analysis; see among others, Reinsel [34], Lütkepohl [28], Tsay [37] and Kilian and Lütkepohl [21].

In this paper, we consider the important case where $q \gg n$ but the VAR(d) model (1) possesses some form of sparse representation, that is many of the parameter coefficients are equal to zero. To elaborate, we first fix some notation. For a random variable X we write $\|X\|_{E,q}$ for $(E|X|^q)^{1/q}$, where $q \in \mathbb{N}$; for a

vector $x \in \mathbb{R}^p$, $\|x\|_0 = \sum_{j=1}^p \mathbb{1}(x_j \neq 0)$, $\|x\|_1 = \sum_{j=1}^p |x_j|$ and $\|x\|_2^2 = \sum_{j=1}^p |x_j|^2$. Furthermore, for a $r \times s$ matrix $B = (b_{i,j})_{i=1,\dots,r,j=1,\dots,s}$, $\|B\|_1 = \max_{1 \leq j \leq s} \sum_{i=1}^r |b_{i,j}| = \max_{1 \leq j \leq s} \|Be_j\|_1$, $\|B\|_\infty = \max_{1 \leq i \leq r} \sum_{j=1}^s |b_{i,j}| = \max_i \|e_i^\top B\|_1$, and $\|B\|_{\max} = \max_{i,j} |e_i^\top Be_j|$, where $e_j = (0, \dots, 0, 1, 0, \dots, 0)^\top$ denotes the vector with the one appearing in the j th position. Denote the largest eigenvalue of a matrix A by $\rho(A)$ and $\|A\|_2^2 = \rho(AA^\top)$. Let $I_{p,-j}$ be the $p \times (p-1)$ matrix obtained from the p dimensional identity matrix after deleting its j th column. Using this notation, let $k_j^r(p) = \sum_{s=1}^d \|e_j^\top A^{(s)}\|_0$ and $k_j^c(p) = \sum_{s=1}^d \|A^{(s)}e_j\|_0$ be the number of non-zero coefficients in the j th row, respectively, in the j th column of the matrices $A^{(s)}$, $s = 1, 2, \dots, d$, and, let $k_1(p) = \max_{1 \leq j \leq p} \{k_j^r(p), k_j^c(p)\}$. In the following we consider the case where the VAR(d) model is sparse, that is the total number of non-zero coefficients within a row or column satisfies $k_1(p) \ll n$. Furthermore, we allow $k_1(p)$ to depend on p , that is, $k_1(p)$ can be an increasing function of the dimension p of the process under consideration. Notice that since a VAR(d) model has a stacked VAR(1) representation (with dimension dp), see Section 2.1, the case of a with sample size increasing order d , can in principle be also covered by our investigations. This is so since an increasing dimension dp of the stacked VAR(1) process can be due to an increasing of p , an increasing of d or an increasing of both parameters. Recall, however, that the effective sample size for a VAR(d) model is $n - d$, see also Wilms et al. [39]. In this paper and in order to simplify notation, we restrict ourselves to the case of a fixed order d .

In the setting described above, procedures to fit sparse VAR models have been considered by many authors in the literature by means of regularized type estimators; see among others, Song and Bickel [36] for ℓ_1 -penalized least squares (lasso) estimators, Han et al. [20] for ℓ_1 -penalized Yule–Walker estimators, Kock and Callot [23] for oracle type inequalities for adaptive lasso estimators and Davis et al. [13] for a two step procedure which includes ℓ_1 -penalized likelihood estimators. Consistency of ℓ_1 -penalized estimators has been established for a specific sparsity setting by Basu and Michailidis [1], while Lin and Michailidis [26] considered estimation for multi-block high-dimensional VAR models including testing of Granger-causality. However, and despite the progress made in fitting sparse VAR models, statistical inference for such models seems to be a less developed area. This is probably due to the fact that the asymptotic distribution of ℓ_1 -penalized estimators is difficult to derive and statistical inference is much more involved and difficult to implement. Notice that even in the i.i.d. regression set-up with fixed dimension, the limiting distribution of regularized lasso estimators has been shown to be nonstandard and one which assigns positive probability mass at zero to the zero coefficients; see Knight and Fu [22]. This leads, among other things, to inconsistency of standard, model-based bootstrap methods; Chatterjee and Lahiri [9]; see also Chatterjee and Lahiri [10] for a different consistent bootstrap proposal in the high-dimensional i.i.d. regression setting.

In this paper, we focus on the development of bootstrap procedures for inferring properties of high-dimensional, sparse VAR(d) processes. Toward this goal and in order to avoid potential problems associated with nonstandard limiting behaviour of regularized lasso estimators, we propose to bootstrap de-biased or de-sparsified estimators of the VAR parameters. De-sparsified estimators of sparse estimators obtained by lasso regularization, have been introduced and investigated in the i.i.d. regression case by several authors. We refer here to the initial paper by Zhang and Zhang [44] and to van de Geer et al. [38], which investigated such estimators in a much more broader setting and established, under certain regularity conditions, asymptotic optimality in the sense of semi-parametric efficiency. In the same i.i.d. regression set-up, de-sparsified estimators have also been used in the context of model based bootstrap inference; see Dezeure et al. [14]. De-biased estimators in the context of Gaussian respectively sub-Gaussian VAR processes have also been considered in Chaudhry et al. [11] and have been used for statistical inference and in particular for testing Granger causality. Furthermore, de-sparsified estimators (or versions of it) were used to do inference for systems of high-dimensional regression equations which included VAR systems. Chernozhukov et al. [12] show a near-to-oracle inequality

for the lasso estimator under a rather general set-up which goes beyond sub-Gaussian innovations. They also do inference by using the bootstrap but they focus on simultaneous equation systems; see in particular Example 2 in Chernozhukov et al. [12]. Thus, the inference set-up considered by these authors does not include VAR processes. Neykov et al. [30] discusses a de-sparsifying approach for Gaussian systems. They construct an influence function by projecting the fitted estimating equations to sparse directions. The resulting de-sparsifying approach is more general in the sense that it can also be applied to nonlinear models. For linear models, like VAR models, estimation of these sparse directions correspond to the CLIME estimator, see Cai et al. [6]. Hence, besides Gaussianity, an important assumption of this approach is that the inverse lag-zero autocovariance matrix of the (stacked) VAR system is sparse. This sparsity is also assumed in Zheng and Raskutti [47] who adapt the de-sparsifying approach of Ning et al. [31] to VAR processes. The innovations are assumed to be componentwise independent, that is, their covariance matrix is diagonal. As we will see in Section 2.1, a sparse VAR system does not necessarily imply that the inverse lag-zero autocovariance matrix is sparse. Hence, such an assumption might be more restrictive in the time series context than it is in the i.i.d. regression set up. We propose here a de-sparsifying approach which makes use of the underlying VAR structure and does not required sparsity of the inverse of the lag-zero autocovariance matrix of the process.

Our work extends and generalizes the aforementioned contributions in many directions. In particular, we consider de-sparsified, respectively, de-biased, estimators for the general VAR(d) process and we derive their limiting distribution under quite general conditions on the process and on the stochastic properties of the, not necessarily Gaussian, innovations. We impose sparsity assumptions only on the VAR coefficient matrices $A^{(s)}$ and on the innovation covariance matrix (or on its inverse), but we *do not* impose sparsity assumptions directly on the autocovariance structure of the VAR(d) process. Furthermore, we introduce a novel and valid bootstrap procedure for inferring properties of the parameters of the VAR(d) process. Appropriately modified, this bootstrap procedure also allows, for testing statistical hypotheses about groups of model parameters in a very flexible way, where the total number of hypotheses tested is also allowed to increase to infinity with the sample size n .

We first derive the limiting distribution of de-sparsified estimators for the parameters of a general, stationary VAR(d) process. We show that this limiting distribution is a regular Gaussian distribution which is solely affected by the autocovariance structure of the underlying VAR(d) process. We then propose a bootstrap procedure to estimate the distribution of the de-sparsified estimators of the VAR parameters. This procedure uses an appropriately thresholded ℓ_1 -penalized estimator of the coefficient matrices $A^{(s)}$ and a thresholded, sparse estimator of the covariance matrix Σ_ε of the innovations. Thresholding is important in this context, since it guaranties sparsity of the VAR model used in the bootstrap world. The fitted sparse VAR model driven by appropriately generated i.i.d. innovations is then used to generate vector pseudo time series $X_1^*, X_2^*, \dots, X_n^*$, which appropriately imitate the sparse stochastic structure of the observed time series $X_1, X_2, \dots, X_n$. We prove consistency of such a bootstrap procedure under general conditions when applied to estimate the distribution of de-sparsified estimators. The results obtained allow for using the bootstrap procedure proposed in order to construct individual or simultaneous confidence intervals and to perform tests of hypotheses about model parameters. In particular, we show how the bootstrap method proposed can be used in order to test the interesting hypothesis that individual or, more importantly, groups of coefficients of the VAR model are equal to zero. For such testing purposes, the bootstrap procedure is appropriately modified so that the sparse VAR model used to generated the pseudo time series $X_1^*, X_2^*, \dots, X_n^*$ satisfies the null hypothesis of interest. Consistency of the bootstrap based testing procedure is established for max-type test statistics. Finally, we demonstrate by means of numerical investigations that the theoretical results established are accompanied by a good finite sample behavior of the bootstrap methodology developed.

The paper is organized as follows. Section 2 introduces de-sparsified estimators for the VAR model parameters and derives their limiting distribution under suitable assumptions on the sparsity of the

underlying VAR process and on the consistency properties of the regularized estimators involved. Section 3 introduces the bootstrap procedure proposed and establishes its asymptotic validity for estimating the distribution of de-sparsified estimators and, appropriately modified, for testing hypotheses about model parameters. Asymptotic validity of the bootstrap based test procedure is established. Section 4 is devoted to issues related to the practical implementation and to numerical investigations. We propose a bias correction procedure to improve the finite sample performance of the bootstrap based testing and we present several simulations supporting the good size and power behavior of the bootstrap methodology proposed for difficult inference problems. An application to an interesting real-life data set is also discussed. Auxiliary results and technical proofs are deferred to Section 6. Finally, some issues related to the theoretical investigations, the numerical simulations and real data applications as well as comparisons with other approaches are discussed in the Supplementary Material to this paper, see Krampe et al. [24].

2. De-sparsified estimators of VAR parameters

2.1. De-sparsified estimators

In this section, we present our approach to construct de-biased estimators in the high-dimensional time series context. Towards this we adapt to the time series set-up the basic idea for the introduction of de-sparsified estimators used in the i.i.d. regression context (see Dezeure et al. [14]) and make the appropriate modifications. We first fix some additional notation. For a matrix $A \in \mathbb{R}^{q \times p}$, $A_{j;r}$ denotes its (j, r) element and $A_{j;-r} := e_j^\top A I_{p;-r}$ the j th row of the matrix A after deleting the element $A_{j;r}$.

Before introducing de-sparsified or de-biased estimators for the general VAR(d) model, we motivate the basic idea of de-biasing in the time series context by first considering the simple case of the VAR(1) model, $X_t = AX_{t-1} + \varepsilon_t$. Recall that if $p \ll n$ the standard least squares estimator of the coefficient $A_{j;r}$ obtained by regressing the j th element of X_t denoted by $X_{t;j}$, on the components of the lagged vector X_{t-1} , can also be written as

$$\hat{A}_{j;r} = \sum_{t=2}^n \tilde{V}_{t-1;r} X_{t;j} / \sum_{t=2}^n \tilde{V}_{t-1;r} X_{t-1;r}. \quad (2)$$

Here $\tilde{V}_{t-1;r}$ denotes the “residuals” obtained as the difference between $X_{t-1;r}$ and its best (in mean square) linear approximation using all other variables contained in the lagged vector X_{t-1} . Notice that for $\Gamma(0) = E(X_t X_t^\top)$ the lag-zero autocovariance matrix, the coefficients of the aforementioned linear approximation can be obtained as $\tilde{\beta}_r = (e_r^\top \tilde{\Gamma}(0)^{-1} e_r)^{-1} \tilde{\Gamma}(0)^{-1} e_r$, where $\tilde{\Gamma}(0)$ is the lag-zero sample autocovariance matrix. Therefore, $\tilde{V}_{t-1;r} = \tilde{\beta}_r^\top X_{t-1}$. Notice that $\tilde{\beta}_r$ is the sample version of $\beta_r = (e_r^\top \Gamma(0)^{-1} e_r)^{-1} \Gamma(0)^{-1} e_r$; see also Lemma 16. Thus, $\hat{A}_{j;r}$ can be interpreted as the effect of the lagged variable $X_{t-1;r}$ on $X_{t;j}$ after eliminating from $X_{t-1;r}$ the (linear) influence of all other variables belonging to the lagged vector X_{t-1} . Clearly, if $p > n$ such an approach to obtain the residuals $\tilde{V}_{t-1;r}$ is not possible. This is due to the fact that in this case, the row vector $(X_{t-1;r}, t = 2, 3, \dots, n)$ is an element of the subspace spanned by the other row vectors $(X_{t-1;s}, t = 2, 3, \dots, n)$ for $s \in \{1, \dots, r-1, r+1, \dots, p\}$, that is if $p > n$ then $\tilde{V}_{t-1;r} \equiv 0$.

In order to overcome this problem in the i.i.d. regression set-up, the approach followed is to replace the estimator for β_r used in the calculation of the residuals $\tilde{V}_{t-1;r}$, by some regularized estimator. This approach requires the imposition of sparsity assumptions on the corresponding vector of coefficients $\beta_r = (e_r^\top \Gamma(0)^{-1} e_r)^{-1} \Gamma(0)^{-1} e_r$, that is on the inverse of the covariance matrix $\Gamma(0)$. However, in the sparse VAR time series set-up considered in this paper, such an approach seems not to be appropriate.

The reason for this lies in the fact that due to temporal dependence, sparsity of the matrix A and Σ_ε respectively Σ_ε^{-1} , does not necessarily imply sparsity of the autocovariance matrices $\Gamma(0)$ respectively $\Gamma(0)^{-1}$. In fact, even in the simple VAR(1) case discussed here, the matrices A and Σ_ε may be sparse but $\Gamma(0)$ respectively $\Gamma(0)^{-1}$ may not. For the same case, observe that if A is symmetric then by properties of the Neumann-series it can be shown that for the inverse of the covariance matrix it yields that $\Gamma(0)^{-1} = \Sigma_\varepsilon^{-1} - \Sigma_\varepsilon^{-1} A \Sigma_\varepsilon A \Sigma_\varepsilon^{-1}$. Therefore, one has to impose certain restrictions on the sparsity of A and Σ_ε , respectively Σ_ε^{-1} , in order to achieve certain sparsity properties of $\Gamma(0)^{-1}$. An example for non-sparse $\Gamma(0)$ and $\Gamma(0)^{-1}$ matrices which appear with sparse A and Σ_ε or Σ_ε^{-1} matrices, respectively, is given in Example 17 in Supplementary Material A.

Thus, in the sparse VAR time series set-up, imposing sparsity assumptions directly on the autocovariance matrices respectively on its inverse, is difficult to justify and may implicitly restrict considerably the class of VAR processes considered. For these reasons, the sparsity behavior of the VAR process is handled in this paper by imposing sparsity assumptions directly on the process parameter matrices $A^{(1)}, \dots, A^{(d)}$ and on Σ_ε , respectively, on Σ_ε^{-1} . Because of this, and in order to obtain appropriate “residuals” $\tilde{V}_{t-1;r}$ and to construct de-sparsified estimators, an alternative approach to those used in the aforementioned discussion has to be followed.

The idea of the procedure we propose in this paper is to estimate the “residuals” needed in the displayed expression for $\hat{A}_{j;r}$, by using the basic expression $\beta_r = (e_r^\top \Gamma(0)^{-1} e_r)^{-1} \Gamma(0)^{-1} e_r$ for the projection coefficients, and by replacing the autocovariance matrix $\Gamma(0)$ by an estimator which is consistent in the high dimensional case ($p > n$). Such a consistent estimator of $\Gamma(0)$ can be obtained by using the well-known relation $\Gamma(0) = \sum_{j=0}^{\infty} A^j \Sigma_\varepsilon (A^\top)^j$ and by replacing A and Σ_ε in this expression by sparse and consistent estimators. One basic problem which occurs in the high dimensional setting in using “residuals” as $\tilde{V}_{t-1;r}$ in (2), is that, due to the fact that the new “residuals” calculated via this approach are not exactly orthogonal to the other variables appearing in the lagged vector $X_{t-1;r}$, a bias term in the corresponding estimator (2) appears. However, this bias term can be consistently estimated which leads to a bias-corrected or de-biased estimator for $A_{j;r}$. In the following, we make these ideas precise by considering the case of the general VAR(d) model.

To start with, recall that the VAR(d) process given in (1) can be written as

$$X_t = (A^{(1)}, \dots, A^{(d)}) (X_{t-1}^\top, \dots, X_{t-d}^\top)^\top + \varepsilon_t =: \Xi W_{t-1} + \varepsilon_t,$$

with an obvious notation for the matrix $\Xi \in \mathbb{R}^{p \times dp}$ and the vector $W_{t-1} \in \mathbb{R}^{dp \times 1}$.

For the VAR(d) process and as already mentioned, if $p \ll n$, the standard least squares estimator of $\Xi_{j;r}$ obtained by regressing $X_{t;j}$ onto $W_{t-1;r}$, $r = 1, \dots, dp$, can also be written as

$$\hat{\Xi}_{j;r} = \sum_{t=d+1}^n \tilde{V}_{t-1;r} X_{t;j} \Big/ \sum_{t=d+1}^n \tilde{V}_{t-1;r} W_{t-1;r}, \quad (3)$$

where $\tilde{V}_{t-1;r}$ is given by $\tilde{V}_{t-1;r} = \tilde{\beta}_r^\top W_{t-1}$, with $\tilde{\beta}_r = (e_r^\top \tilde{\Gamma}^{(st)}(0)^{-1} e_r)^{-1} \tilde{\Gamma}^{(st)}(0)^{-1} e_r$ and $\tilde{\Gamma}^{(st)}(0)$ the sample covariance matrix of $\{W_t, t = d+1, \dots, n\}$. Notice that $\tilde{\Gamma}^{(st)}(0) = (n-d)^{-1} \sum_{t=d+1}^n (W_t - \bar{W}_n)(W_t - \bar{W}_n)^\top$ and $\bar{W}_n = (n-d)^{-1} \sum_{t=d+1}^n W_t$. Furthermore, $\tilde{\beta}_r$ is an estimator of $\beta_r = (e_r^\top \Gamma^{(st)}(0)^{-1} e_r)^{-1} \Gamma^{(st)}(0)^{-1} e_r$. If $p > n$, then $\tilde{\Gamma}^{(st)}(0)$ is not a consistent estimation of $\Gamma^{(st)}(0)$ (or of its inverse). Thus, a suitable estimator of $\Gamma^{(st)}(0)$ has to be used in the high dimensional setting which satisfies certain consistency properties. We will discuss later on the construction of such an estimator based on the generalization to VAR(d) processes of the expression $\Gamma(0) = \sum_{j=0}^{\infty} A^j \Sigma_\varepsilon (A^\top)^j$. For the moment, denote such an estimator by $\hat{\Gamma}^{(st)}(0)$. We can then construct our “residuals” which are denoted by $\hat{Z}_{t-1;r}$, as $\hat{Z}_{t-1;r} = \hat{\beta}_r^\top W_{t-1}$, where $\hat{\beta}_r$ is defined by $\hat{\beta}_r = (e_r^\top \hat{\Gamma}^{(st)}(0)^{-1} e_r)^{-1} \hat{\Gamma}^{(st)}(0)^{-1} e_r$.

We then have

$$\begin{aligned}\tilde{\Xi}_{j;r} &= \sum_{t=d+1}^n \hat{Z}_{t-1;r} X_{t;j} / \sum_{t=d+1}^n \hat{Z}_{t-1;r} W_{t-1;r} \\ &= \Xi_{j;r} + \sum_{t=d+1}^n \hat{Z}_{t-1;r} \varepsilon_{t;j} / \sum_{t=d+1}^n \hat{Z}_{t-1;r} W_{t-1;r} \\ &\quad + \sum_{t=d+1}^n \hat{Z}_{t-1;r} W_{t-1;-r} \Xi^\top e_j / \sum_{t=d+1}^n \hat{Z}_{t-1;r} W_{t-1;r}.\end{aligned}$$

Since $p > n$, the “residuals” $\{Z_{t-1;r}\}$ are not perfectly orthogonal to $\{W_{t-1;-r}\}$. Therefore, the last term of the last equation above is non-zero, which is the bias that the estimator $\tilde{\Xi}_{j;r}$ produces. However, this bias can be estimated using some (regularized) estimator of $\Xi = (A^{(1)}, \dots, A^{(d)})$ that satisfies certain consistency properties. We will discuss such an estimator later on. Given such an estimator, which we denote by $\hat{\Xi}^{(\text{re})} = (\hat{A}^{(1,\text{re})}, \dots, \hat{A}^{(d,\text{re})})$, we can estimate the bias term in the last displayed expression by replacing Ξ by $\hat{\Xi}^{(\text{re})}$. Thus, substituting $\hat{\Xi}^{(\text{re})}$ for Ξ , subtracting the estimated bias from $\tilde{\Xi}_{j;r}$ and rearranging terms, we get the following de-biased estimator of $\Xi_{j;r}$,

$$\hat{\Xi}_{j;r}^{(\text{de})} = \hat{\Xi}_{j;r}^{(\text{re})} + \left(\sum_{t=d+1}^n \hat{Z}_{t-1;r} W_{t-1;r} \right)^{-1} \left(\sum_{t=d+1}^n \hat{Z}_{t-1;r} (X_{t;j} - e_j^\top \hat{\Xi}^{(\text{re})} W_{t-1}) \right). \quad (4)$$

The estimator $\hat{\Xi}_{j;r}^{(\text{de})}$ given above is called a de-biased, respectively, a de-sparsified estimator of $\Xi_{j;r}$ since it is a bias corrected version of the initial estimator $\tilde{\Xi}_{j;r}$ and it is not sparse anymore. Note that $A_{j;r}^{(s)} = \Xi_{j;(s-1)p+r}$ and, therefore, the corresponding de-sparsified estimator of $A_{j;r}$ is given by $\hat{A}_{j;r}^{(s,\text{de})} = \hat{\Xi}_{j;(s-1)p+r}^{(\text{de})}$ for $j, r = 1, \dots, p$ and $s = 1, \dots, d$.

We now discuss the estimators $\hat{\Xi}^{(\text{re})}$ and $\hat{\Gamma}^{(st)}(0)$ used in the above construction. Suitable candidates for $\hat{\Xi}^{(\text{re})} = (\hat{A}^{(1,\text{re})}, \dots, \hat{A}^{(d,\text{re})})$ in the sparse VAR(d) setting are the adaptive lasso estimator investigated by Kock and Callot [23]; see also Chernozhukov et al. [12] and the discussion following equation (7) below. Such an estimator for the i th row of Ξ denoted by $\hat{\Xi}_{i,\cdot}^{(\text{re})}$, is obtained as

$$\hat{\Xi}_{i,\cdot}^{(\text{re})} = \underset{c=(c_1, \dots, c_{dp})^\top \in \mathbb{R}^{dp}}{\operatorname{argmin}} \frac{1}{n-d} \sum_{t=d+1}^n (X_{i;t} - c^\top W_{t-1})^2 + \lambda_n \sum_{s=1}^{dp} \frac{|c_s|}{|\hat{c}_{j,s}|}, \quad (5)$$

where $\hat{c}_j = (\hat{c}_{j,s}, s = 1, 2, \dots, dp)$ are the lasso estimators of $\Xi_{i,\cdot}$ obtained as $\hat{c}_j = \operatorname{argmin}_{c \in \mathbb{R}^{dp}} (n-d)^{-1} \sum_{t=d+1}^n (X_{i;t} - c^\top W_{t-1})^2 + \lambda_n \|c\|_1$ and $\lambda_n = C \sqrt{\log(p)/n}$ is a regularization parameter.

Estimation of the covariance matrix $\Gamma^{(st)}(0)$ is more involved. As already mentioned, for a stable VAR(1) process with coefficient matrix $A = A^{(1)}$ we have $\Gamma(0) = \sum_{j=0}^{\infty} A^j \Sigma_\varepsilon (A^\top)^j = \operatorname{vec}^{-1}(I_{p^2} - A \otimes A)^{-1} \operatorname{vec}(\Sigma_\varepsilon)$. This expression can be generalized to a VAR(d) process using its stacked VAR(1) representation, that is, the representation $W_t = \mathbb{A} W_{t-1} + \mathbb{U}_t$; see Supplementary Material A for the definition of $\mathbb{A}$ and $\mathbb{U}$. This representation leads to the expression

$$\Gamma^{(st)}(0) = \sum_{j=0}^{\infty} \mathbb{A}^j \Sigma_{\mathbb{U}} (\mathbb{A}^\top)^j = \operatorname{vec}^{-1}((I_{(dp)^2} - \mathbb{A} \otimes \mathbb{A})^{-1} \operatorname{vec}(\Sigma_{\mathbb{U}})). \quad (6)$$

The above expression suggests that a consistent estimator $\widehat{\Gamma}^{(st)}(0)$ of $\Gamma^{(st)}(0)$ can be obtained by plugging in (6) estimators $\widehat{A}^{(s)}$, $s = 1, 2, \dots, d$ and $\widehat{\Sigma}_\varepsilon$ of the coefficient matrices $A^{(s)}$, $s = 1, 2, \dots, d$ and of the covariance matrix Σ_ε , respectively, provided these estimators satisfy certain consistency properties, see Lemma 8 for details.

To elaborate on the construction of the estimator $\widehat{A}^{(s)}$, as expression (6) shows, $\|\cdot\|_\infty$ consistency of $\widehat{\Gamma}^{(st)}(0)$ at a desired rate, requires consistency at an appropriate rate of $\sum_{s=1}^p \|\widehat{A}^{(s)} - A^{(s)}\|_\infty = \sum_{s=1}^p \|\widehat{A}^{(s)} - A^{(s)}\|_1$ and of $\sum_{s=1}^p \|\widehat{A}^{(s)} - A^{(s)}\|_\infty$. That is, we need to control the errors made in estimating the coefficients $A_{j;r}^{(s)}$ in the rows and in the columns of the matrices $A^{(s)}$, $s = 1, \dots, d$, taking at the same time into account that the dimension of these matrices is allowed to increase to infinity with n . These considerations motivate the following estimator of $A_{j;r}^{(s)}$:

$$\widehat{A}_{j;r}^{(s)} = \widehat{A}_{j;r}^{(s,\text{re})} \mathbb{1}(|\widehat{A}_{j;r}^{(s,\text{re})}| \geq a_n), \quad j, r = 1, \dots, p, s = 1, \dots, d, \quad (7)$$

where $\widehat{A}_{j;r}^{(s,\text{re})}$ is some initial estimator and a_n a threshold parameter; we refer to Lemma 9 for details and properties of this estimator. Suitable candidates for $\widehat{A}_{j;r}^{(s,\text{re})}$ are the adaptive lasso estimator (5), see also Kock and Callot [23] and the regularized Yule–Walker type estimator (8), see Han et al. [20]; Wu et al. [40]. Notice that the regularized Yule–Walker estimator of the i th row of the matrix Ξ , denote by $\widehat{\Xi}_{i,\cdot}^{(YW)}$, is obtained as

$$\widehat{\Xi}_{i,\cdot}^{(YW)} = \underset{c \in \mathbb{R}^{dp}}{\operatorname{argmin}} \{ \|c\|_1 \text{ s.t. } \|\widehat{\Gamma}_W(0)c - \widehat{\Gamma}_W(1)e_i\|_\infty \leq \lambda_n \}, \quad (8)$$

where $\widehat{\Gamma}_W(0) = (n-d)^{-1} \sum_{t=d}^n W_t W_t^\top$, $\widehat{\Gamma}_W(1) = (n-d)^{-1} \sum_{t=d}^{n-1} W_t W_{t+1}^\top$ and $\lambda_n > 0$ is a regularization parameter. To further elaborate on the motivation leading to the estimator (7), observe that similar to covariance estimation, thresholding transfers element-wise consistency to consistency results with respect to a matrix norm. In particular, Kock and Callot [23] obtained under some conditions which include Gaussian innovations, that the row-wise ℓ_1 -error of the adaptive lasso has the rate $O_P(k_1(p)\sqrt{(\log(n)\log(p)\log(d))^5} \times 1/\sqrt{n})$. Furthermore, Basu and Michailidis [1] obtained under some conditions which include Gaussian innovations in a stochastic regression setting similar to (5), an ℓ_2 -error bound with rate $O_P(\sqrt{k_1(p)\log(pd)/n})$. This gives an upper bound for the elementary-wise error. Han et al. [20] showed under some conditions which also include Gaussian innovations, that $\|\widehat{\Xi}^{(YW)} - \Xi\|_{\max} = O_P(\|\Xi\|_1 \|\Gamma^{(st)}(0)^{-1}\|_1 \sqrt{\log(p)/n})$.

To estimate the innovation covariance matrix Σ_ε , several approaches exist which depend on the sparsity assumptions one wants to impose on this matrix; see Cai et al. [8] for a discussion in the i.i.d. setting. If one imposes sparsity assumptions on Σ_ε then Bickel and Levina [3], El Karoui [17], Cai and Liu [5], see also Lemma 10, provide some thresholding-based approaches for estimating Σ_ε . If sparsity assumptions are imposed on Σ_ε^{-1} , we refer to Cai et al. [7]. If one solely requires that $p/n \rightarrow c \in (0, 1)$, then Ledoit and Wolf [25] present approaches to estimate the corresponding covariance matrix.

We elaborate here on estimating Σ_ε when sparsity assumptions are imposed on Σ_ε . Let $\widehat{\varepsilon}_t = X_t - \sum_{s=1}^d \widehat{A}^{(s)} X_{t-s}$, $t = d+1, \dots, n$, be the estimated residuals of the VAR fit and assume that $\max_j \|\Sigma_\varepsilon e_j\|_0 = O(k_2(p))$ and $\|\Sigma_\varepsilon\|_1 = O(k_2(p))$. For $\bar{\varepsilon}_n = 1/(n-d) \sum_{t=d+1}^n \widehat{\varepsilon}_t$ let $\widetilde{\Sigma}_\varepsilon = (n-d)^{-1} \sum_{t=d+1}^n (\widehat{\varepsilon}_t - \bar{\varepsilon}_n)(\widehat{\varepsilon}_t - \bar{\varepsilon}_n)^\top$ and

$$\widehat{\Sigma}_\varepsilon = (\widetilde{\Sigma}_{\varepsilon,i,j} \mathbb{1}\{|\widetilde{\Sigma}_{\varepsilon,i,j}| \geq b_n\})_{i,j=1,2,\dots,p}, \quad (9)$$

where $\widetilde{\Sigma}_{\varepsilon,i,j}$ denotes the (i, j) th element of $\widetilde{\Sigma}_\varepsilon$; we refer to Lemma 10 for properties of the estimator $\widehat{\Sigma}_\varepsilon$ and more specifically that this estimator satisfies Assumption 1(iv) of the next section. In particular,

it is shown in this lemma, that under the assumption of Gaussian innovations ε_t , this estimator satisfies that $\|\widehat{\Sigma}_\varepsilon - \Sigma_\varepsilon\|_1 = O_P(k_2(p)\sqrt{\log(p)/n})$. The results of Bickel and Levina [3], Section 2.3 and El Karoui [17], indicate that for the non-Gaussian case slower rates should be expected. However, Cai and Liu [5] showed that with a more refined thresholding strategy the same rate also can be obtained in the non-Gaussian i.i.d. case. They suggest using individual thresholding values instead of applying an universal thresholding parameter. A simple modification of (9) taking their considerations into account leads to $\widehat{\Sigma}_\varepsilon = (\widetilde{\Sigma}_{\varepsilon,i,j} \mathbb{1}\{| \widetilde{R}_{\varepsilon,i,j} | \geq b_n\})_{i,j=1,\dots,p}$ where $\widetilde{R}_{\varepsilon,i,j} = \widetilde{\Sigma}_{\varepsilon,i,j} / (\widetilde{\Sigma}_{\varepsilon,i,i} \widetilde{\Sigma}_{\varepsilon,j,j})^{1/2}$.

Note that if the estimators $\widehat{A}^{(s)}$ are such that the matrix polynomial $\widehat{A}(z) = I - \sum_{s=1}^d \widehat{A}^{(s)} z^s$ has all its roots outside the unit disc, that is, $\rho(\widehat{\mathbb{A}}) < 1$, and if $\widehat{\Sigma}_\varepsilon$ is positive definite, then $\widehat{\Gamma}^{(st)}(0)$ is positive definite and the estimator $\widehat{\beta}_r = \widehat{\Gamma}^{(st)}(0)^{-1} e_r / [e_r^\top \widehat{\Gamma}^{(st)}(0)^{-1} e_r]$ used in the calculation of the residuals $\widehat{Z}_{t-1;r}$ is well defined.

2.2. Asymptotic distribution of de-sparsified estimators

To derive the asymptotic distribution of $\widehat{\Xi}_{j;r}^{(\text{de})}$ we first observe that by substituting expression $X_{t;j} = e_j^\top \Xi W_{t-1} + \varepsilon_{t;j}$ in (4), that this estimator can be written as,

$$\widehat{\Xi}_{j;r}^{(\text{de})} = \Xi_{j;r} + \left[\sum_{t=d+1}^n \widehat{Z}_{t-1;r} \varepsilon_{t;j} + \widehat{Z}_{t-1;r} W_{t-1;-r} (\Xi_{j;-r} - \widehat{\Xi}_{j;-r}^{(\text{re})}) \right] / \sum_{t=d+1}^n \widehat{Z}_{t-1;r} W_{t-1;r}. \quad (10)$$

Representation (10) suggests that asymptotic normality of the de-sparsified estimator, more precisely of $\sqrt{n}(\widehat{A}_{j;r}^{(s,\text{de})} - A_{j;r}^{(s)})$ can be established, due to the contribution of $\sum_{t=d+1}^n \widehat{Z}_{t-1;r} \varepsilon_{t;j}$ on the right-hand side of (10) and the asymptotic negligibility of the second term which is due to the estimation error $\Xi_{j;-((s-1)p+r)} - \widehat{\Xi}_{j;-((s-1)p+r)}^{(\text{re})}$.

Theorem 1 below confirms that this intuition is indeed true. However, to state precisely this theorem, we impose some conditions on the underlying VAR process, on its sparsity as well as on the consistency properties of the estimators involved in the construction of the de-sparsified estimator $\widehat{A}_{j;r}^{(s,\text{de})} = \widehat{\Xi}_{j;(s-1)p+r}^{(\text{de})}$.

Assumption 1.

(i)

$$\max_{1 \leq i \leq p} \sum_{s=1}^d \|e_i^\top A^{(s)}\|_0 = O(k_1(p)), \quad \max_{1 \leq i \leq p} \sum_{s=1}^d \|A^{(s)} e_i\|_0 = O(k_1(p)) \quad \text{and} \\ \max_{1 \leq i \leq p} \|e_i^\top \Sigma_\varepsilon\|_0 = O(k_2(p)).$$

(ii) There exists a $\lambda \in (0, 1)$ such that $\rho(\mathbb{A}) \leq \lambda$ and for any $k \in \mathbb{N}$,

$$\|\mathbb{A}^k\|_2 = O(\lambda^k), \quad \|\mathbb{A}^k\|_1 = O(k_1(p)\lambda^k) \quad \text{and} \quad \|\mathbb{A}^k\|_\infty = O(k_1(p)\lambda^k).$$

(iii) The estimator $\widehat{\Xi}^{(\text{re})} = (\widehat{A}^{(1,\text{re})}, \dots, \widehat{A}^{(d,\text{re})})$ used in estimating the bias term in (4) satisfies

$$\sum_{s=1}^d \|\widehat{A}^{(s,\text{re})} - A^{(s)}\|_{\infty} = O_P(k_1(p)\sqrt{g(p, d, n)/n})$$

while the estimator $\widehat{A}^{(s)}$ given in (7) and used for estimating $\widehat{\Gamma}^{(st)}(0)$, satisfies

$$\sum_{s=1}^d \|\widehat{A}^{(s)} - A^{(s)}\|_l = O_P(k_1(p)^{3/2}\sqrt{g(p, d, n)/n}), \quad l \in \{1, \infty\}.$$

(iv) $\|\Sigma_{\varepsilon} - \widehat{\Sigma}_{\varepsilon}\|_1 = O_P(k_2(p)\sqrt{g(p, d, n)/n})$.

(v) $\|\Gamma^{(st)}(0)^{-1}\|_{\infty} = O(k_3(p))$.

(vi) $(E(v^{\top} \varepsilon_0)^q)^{1/q} < \infty$ for all $\|v\|_2 < \infty$ for some $q \geq 8$ such that

$$p/(n^{q/4-1} \log^{q/4}(p)) = O(1) \quad \text{and} \quad k_1(p)^6 k_2(p) k_3(p)^2 g(p, d, n)/\sqrt{n} = o(1).$$

Notice that g appearing in the above expressions is an increasing function of the dimension p which is allowed to be different from expression to expression.

Some comments on the above assumptions are in order. The first two statements of Assumption 1(i) refer to the sparsity behavior of the coefficient matrices $A^{(s)}$. According to this assumption, $k_1(p)$ is the maximum number of non-zero coefficients that can appear in each row, respectively, in each column of the matrices $A^{(s)}$. This implies that $\sum_{s=1}^d \|\text{vec}(A^{(s)})\|_0 = O(k_1(p)p)$ which is a more flexible sparsity setting for VAR(d) models compared, for instance, to the one used in Basu and Michailidis [1] and which requires that $\sum_{s=1}^d \|\text{vec}(A^{(s)})\|_0 \leq k$. To elaborate, under the aforementioned sparsity requirement $\sum_{s=1}^d \|\text{vec}(A^{(s)})\|_0 = k$ and the assumption of Gaussian innovations, Basu and Michailidis [1] derived the error bound $O_P(k\sqrt{\log(p)/n})$ with respect to the $\|\text{vec}(\cdot)\|_1$ norm for their estimator. Consistency to $\|\text{vec}(\cdot)\|_1$ implies consistency to $\|\cdot\|_1$ and $\|\cdot\|_{\infty}$ but not vice versa. Furthermore, the sparsity setting $\sum_{s=1}^d \|\text{vec}(A^{(s)})\|_0 = k$ essentially implies that if non i.i.d. processes are added to the system, the dimension p is allowed to increase at a rate which is smaller than $\sqrt{n}$. In contrast to this, under our sparsity assumptions, and if we allow for Gaussian innovations, then any polynomial growth rate of p with respect to the sample size n is allowed; see also Supplementary Material F for a more detailed comparison of the sparsity setting used here and in Basu and Michailidis [1]. The third statement of Assumption 1(i) refers to the sparsity of the innovation covariance matrix Σ_{ε} . They have to be modified appropriately if one prefers to put sparsity assumptions on the inverse matrix $\Sigma_{\varepsilon}^{-1}$.

Assumption 1(ii) imply that the VAR model considered is stable and further satisfies some kind of uniformity regarding the decay behavior of the coefficient matrices A^k . This assumption seems necessary because the dimension p of the process $\{X_t, t \in \mathbb{Z}\}$ is allowed to increase to infinity with n . To elaborate, let for simplicity $d = 1$. Then it is well known that $A^k = B_k$, where B_k are the coefficient matrices appearing in the infinite order, causal moving average representation $X_t = \sum_{k=1}^{\infty} B_k \varepsilon_{t-k} + \varepsilon_t$ of X_t . Furthermore, these coefficients decrease exponentially fast to zero, that is $\|A^k\|_j \leq C\rho^k$, for $j \in \{1, \infty\}$ and for some constants C and $\rho \in (0, 1)$, which will eventually vary when the dimension of the process changes. This situation is taking care off in Assumption 1(ii) which controls the decay rate of A^k by the same constant $\lambda \in (0, 1)$.

Assumption 1(iii) states the required row wise consistency rates for the estimator $\widehat{A}^{(s,\text{re})}$ and the row- and column wise consistency rates of $\widehat{A}^{(s)}$. Notice first that as this assumption shows, we can choose

$\widehat{A}^{(s, \text{re})}$ to be the same estimator as $\widehat{A}^{(s)}$, $s = 1, 2, \dots, d$, where the latter estimator is given in (7). However, the thresholding used in (7) may lead to an additional bias in finite samples. This is why we prefer to distinguish the two estimators $\widehat{A}^{(s)}$ and $\widehat{A}^{(s, \text{re})}$, which are used for different purposes. Furthermore, and as we have already mentioned in the discussion following equation (7), under Gaussian assumptions on the innovations ε_t , the estimator $\widehat{A}^{(s)}$ satisfies by construction the required consistency rates with respect to both norms. Observe that we left g in Assumption 1(iii) unspecified since its particular form depends, among other things, also on the distribution of the i.i.d. innovations. As already mentioned, for Gaussian innovations, Kock and Callot [23] showed under standard lasso conditions that the adaptive lasso estimator of $A^{(s)}$ satisfy the rates of Assumption 1(iii) with the function $g(p, d, n)$ given by $g(p, d, n) = (\log(n) \log(p) \log(d))^5$. Similar results hold also for the sub-Gaussian case, see Zheng and Raskutti [47] and Chernozhukov et al. [12]. If the distribution of the i.i.d. innovations has only a limited number of finite moments, Chernozhukov et al. [12] pointed out that g drops to a polynomial rate; see among others Comment 5.4 of the aforementioned paper.

Assumption 1(iv) refers to the consistency rates of the estimator $\widehat{\Sigma}_\varepsilon$. Notice that as Lemma 10 shows, the thresholded estimator $\widehat{\Sigma}_\varepsilon$ given in (9) satisfies under the imposed sparsity assumptions and for Gaussian innovations the consistency condition stated in this assumption. Assumption 1(v) controls the growth rates of the inverse of the stacked covariance matrix $\Gamma^{(st)}(0)$. Note that the stacked covariance itself is affected by the increasing number of non-zero elements $k_1(p)$ and $k_2(p)$ but the inverse $\Gamma^{(st)}(0)^{-1}$ is also affected by the smallest eigenvalue of $\Gamma^{(st)}(0)$. Assumption 1(vi) imposes a moment condition on the innovations ε_t and specifies its effect on the allowed growth rates of the dimension p and of the sparsity behavior of the VAR process when n increases. For instance, and as already mentioned, if Gaussian innovations are assumed, then any polynomial growth rate of p in relation to n is allowed.

We now state a result which establishes asymptotic normality of the de-sparsified estimators given in (4).

Theorem 1. *Suppose that Assumption 1 is satisfied. Then for all $j, r \in \{1, \dots, p\}$ and $s = 1, \dots, d$, we have, as $n \rightarrow \infty$,*

$$\sqrt{n}(\widehat{A}_{j;r}^{(s, \text{de})} - A_{j;r}^{(s)}) \xrightarrow{D} \mathcal{N}(0, s.e.(j, r, s)^2), \quad (11)$$

where $s.e.(j, r, s)^2 = \Sigma_{\varepsilon, j; j} (e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} e_{(s-1)p+r})$.

The following lemma deals with the limiting covariance of the same estimators.

Lemma 2. *Under the conditions of Theorem 1 we have for any $s_1, s_2 \in \{1, \dots, d\}$ and $j_1, j_2, r_1, r_2 \in \{1, \dots, p\}$, that, as $n \rightarrow \infty$,*

$$\text{Cov}(\sqrt{n}\widehat{A}_{j_1;r_1}^{(s_1, \text{de})}, \sqrt{n}\widehat{A}_{j_2;r_2}^{(s_2, \text{de})}) \rightarrow \Sigma_{\varepsilon, j_1; j_2} e_{(s_1-1)p+r_1}^\top (\Gamma^{(st)}(0))^{-1} e_{(s_2-1)p+r_2}. \quad (12)$$

3. Bootstrap based inference for sparse vector autoregressions

3.1. Bootstrapping de-sparsified estimators

Due to their regular limiting distribution, de-sparsified estimators can be used as a vehicle for statistical inference for sparse VAR(d) models. In this section, we introduce a bootstrap procedure for this purpose. Note that in contrast to the i.i.d. regression setting, in the VAR(d) setting considered in this

paper, the “regressors” $(X_{t-1}, \dots, X_{t-d}, t = d+1, \dots, n)$, that is, the vectors $\{W_{t-1}, t = d+1, \dots, n\}$ are not fixed; they are dependent random vectors. In order to capture the randomness and the dependence properties of these vectors, an appropriate bootstrap procedure has to be applied, which uses an estimated and sparsified version of the underlying $\text{VAR}(d)$ model to generate pseudo time series $X_1^*, X_2^*, \dots, X_n^*$. This pseudo time series can then be used to infer properties of $A_{j;r}^{(s)}$ using the estimated distribution of $\sqrt{n}(\widehat{A}_{j;r}^{(s, \text{de})} - A_{j;r}^{(s)})$.

A standard procedure to bootstrap a VAR process in the finite, low dimensional case, is to estimate the model and to generate pseudo time series by using the estimated model structure and by drawing with replacement from the estimated set of residuals. However, this procedure fails in the high-dimensional case. To elaborate, let $\widehat{\varepsilon}_t$ be the centered, estimated residuals of a VAR fit. If the pseudo innovations ε_t^* are obtained by drawing with replacement from the set $\{\widehat{\varepsilon}_t, t = d+1, \dots, n\}$, then $\text{Var}^* \varepsilon_t^* = 1/(n-d) \sum_{t=d+1}^n \widehat{\varepsilon}_t \widehat{\varepsilon}_t^\top = \widetilde{\Sigma}_\varepsilon$. However, it is well known that the sample covariance matrix $\widetilde{\Sigma}_\varepsilon$ is not a valid estimator of Σ_ε in the high-dimensional case. Thus, such a bootstrap approach would fail to appropriately mimic the second-order properties of the time series at hand. To tackle this problem, a different strategy is proposed. To elaborate, note first that as Theorem 1 shows, the limiting distribution of the de-sparsified estimators is not affected by the distribution of the ε_t ’s but solely by their second-order properties, that is by the covariance matrix Σ_ε . Thus, we generate the pseudo innovations $\{\varepsilon_t^*\}$ as Gaussian random vectors with covariance matrix $\widehat{\Sigma}_\varepsilon$. The estimator $\widehat{\Sigma}_\varepsilon$ is consistent in the high-dimensional setting which ensures that $\{\varepsilon_t^*\}$ generated in this way will correctly mimic the second-order properties of the true innovations $\{\varepsilon_t\}$. Notice that instead of the Gaussian distribution any other distribution could be used as well, provided this distribution possesses at least q moments. Furthermore, using the Gaussian distribution to generate the pseudo innovations, does not affect the asymptotic validity of the bootstrap procedure proposed in the sequel, when this procedure is applied to estimate the distribution of the de-sparsified estimators and to perform tests for groups of coefficients.

Apart from using the Gaussian distribution with covariance matrix $\widehat{\Sigma}_\varepsilon$ to generate the pseudo innovations, the thresholded estimators $\widehat{A}^{(s)}$ given in (7) are used to generate the pseudo time series. This enables the bootstrap generated pseudo time series $X_1^*, X_2^*, \dots, X_n^*$ to appropriately mimic the dependence and the sparsity properties of the underlying $\text{VAR}(d)$ model. Note that thresholding is important in the sparse high dimensional context since it guarantees sparsity of the VAR model used in the bootstrap world. Using the aforementioned estimators $\widehat{A}^{(s)}$ and $\widehat{\Sigma}_\varepsilon$, the generated pseudo time series $X_1^*, X_2^*, \dots, X_n^*$ stems from a $\text{VAR}(d)$ model which has $\widehat{\Gamma}^{(st)}(0)$ as its lag-zero autocovariance matrix. Recall that the estimator $\widehat{A}^{(s)}$ given in (7), selects the non-zero components of the matrices $A^{(s)}$ by means of thresholding an initial estimator. Notice that, as we will see, consistency of this estimator as well as a bounded row- and column-wise support property is sufficient for validity of the bootstrap procedure proposed. More specifically, and as a careful inspection of the proof of Theorem 3 below shows, a full sign recovery, which will require a minimal signal strength condition on the coefficients $A_{j;r}^{(s)}$, is not needed. Notice that under such an additional minimal signal strength condition, consistent estimation of the support of $A^{(s)}$, $s = 1, \dots, d$, also can be established; see Theorem 7 of Kock and Callot [23] and Corollary 1 of Han et al. [20]. Lack of such a minimal signal strength condition does not affect the consistency and the aforementioned row- and columnwise boundedness properties of the estimators $\widehat{A}^{(s)}$, which are used for the bootstrap.

The bootstrap algorithm proposed consists of the following four steps.

Step 1: Generate i.i.d. pseudo innovations ε_t^* from $\mathcal{N}(0, \widehat{\Sigma}_\varepsilon)$.

Step 2: Generate a pseudo time series $X_1^*, X_2^*, \dots, X_n^*$ using the model equation

$$X_t^* = \sum_{s=1}^d \widehat{A}^{(s)} X_{t-s}^* + \varepsilon_t^*, \quad t = 1, 2, \dots, n$$

and some starting values $X_0^*, X_{-1}^*, \dots, X_{1-d}^*$.

Step 3: Let $\widehat{A}_{j;r}^{*(s,de)}$ be the same de-sparsified estimator of $A_{j;r}^{(s)}$ as the estimator $\widehat{A}_{j;r}^{(s,de)}$ given in (4), but based on the pseudo time series $X_1^*, X_2^*, \dots, X_n^*$.

Step 4: Approximate the distribution of $\sqrt{n}(\widehat{A}_{j;r}^{(s,de)} - A_{j;r}^{(s)})$ by that of the bootstrap analogue $\sqrt{n}(\widehat{A}_{j;r}^{*(s,de)} - \widehat{A}_{j;r}^{(s)})$.

Validity of the bootstrap procedure in approximating consistently the distribution of de-sparsified estimators is established in the following theorem, where Mallows's d_2 metric is used to measure the distance between two distributions. For two random variables X and Y , Mallows's distance between their distributions is defined as $d_2(X, Y) = \{\int_0^1 (F_X^{-1}(x) - F_Y^{-1}(x))^2 dx\}^{1/2}$ where F_X and F_Y denote the cumulative distribution functions of X and Y , respectively; see Bickel and Freedman [2].

Theorem 3. Suppose that Assumption 1 is fulfilled. Then, we have for all $j, r \in \{1, \dots, p\}$ and $s \in \{1, \dots, d\}$, that as $n \rightarrow \infty$,

$$d_2(\sqrt{n}(\widehat{A}_{j;r}^{(s,de)} - A_{j;r}^{(s)}), \sqrt{n}(\widehat{A}_{j;r}^{*(s,de)} - \widehat{A}_{j;r}^{(s)})) = o_P(1). \quad (13)$$

The next corollary establishes validity of the corresponding studentized distributions as well.

Corollary 4. Under the assumptions of Theorem 3 we have, as $n \rightarrow \infty$, that

$$d_2(\sqrt{n}(\widehat{A}_{j;r}^{(s,de)} - A_{j;r}^{(s)})/(\widehat{s.e.}(j, r, s)), \sqrt{n}(\widehat{A}_{j;r}^{*(s,de)} - \widehat{A}_{j;r}^{(s)})/(\widehat{s.e.}^*(j, r, s))) = o_P(1), \quad (14)$$

where

$$\widehat{s.e.}(j, r, s)^2 = \widehat{\Sigma}_{\varepsilon, j; j} (e_{(s-1)p+r}^\top (\widehat{\Gamma}^{(st)}(0))^{-1} e_{(s-1)p+r}) \quad (15)$$

and

$$\widehat{s.e.}^*(j, r, s)^2 = \widehat{\Sigma}_{\varepsilon, j; j}^* ((e_{(s-1)p+r}^\top (\widehat{\Gamma}^{*(st)}(0))^{-1} e_{(s-1)p+r}). \quad (16)$$

Here $\widehat{\Sigma}_{\varepsilon, j; j}^*$ and $\widehat{\Gamma}^{*(st)}(0)$ denote the same quantities as $\widehat{\Sigma}_{\varepsilon, j; j}$ and $\widehat{\Gamma}^{(st)}(0)$ but based on the bootstrap pseudo time series $X_1^*, X_2^*, \dots, X_n^*$. The estimator $\widehat{\Sigma}_{\varepsilon, j; j}^*$ is the j th diagonal entry of the estimator $\widehat{\Sigma}_\varepsilon$ used in Step 1 of the bootstrap algorithm.

The last result of this section shows that the bootstrap version of the de-sparsified estimators considered, can also be used to consistently estimate the covariance of the de-sparsified estimators of the coefficients $A_{j_1; r_1}^{(s_1)}$ and $A_{j_2; r_2}^{(s_2)}$. Its proof follows along the same lines as the proof of Lemma 2 and uses arguments similar to those applied in the proof of Theorem 3.

Lemma 5. *Under the assumptions of Theorem 3 we have for any $s_1, s_2 \in \{1, \dots, d\}$ and $j_1, j_2, r_1, r_2 \in \{1, \dots, p\}$, that, as $n \rightarrow \infty$,*

$$\text{Cov}^*(\sqrt{n}\hat{A}_{j_1;r_1}^{*(s_1, \text{de})}, \sqrt{n}\hat{A}_{j_2;r_2}^{*(s_2, \text{de})}) \rightarrow \Sigma_{\varepsilon, j_1; j_2} e_{(s_1-1)p+r_1}^\top (\Gamma^{(st)}(0))^{-1} e_{(s_2-1)p+r_2}, \quad (17)$$

in probability.

3.2. Testing statistical hypotheses

Appropriately modified, the bootstrap procedure proposed can also be used in order to test hypotheses of interest regarding the dependence structure of the underlying VAR(d) model, like for instance, testing that a subset of the parameters of the VAR model is zero. To elaborate, let $G \subseteq \{(j, r, s) : j, r \in \{1, \dots, p\} \text{ and } s \in \{1, \dots, d\}\}$ be a subset of indices and consider the following testing problem:

$$H_0: A_{j;r}^{(s)} = 0, \text{ for all } (j, r, s) \in G.$$

$$H_1: \text{There exists at least one } (j, r, s) \in G \text{ such that } A_{j;r}^{(s)} \neq 0.$$

We assume in the following that the restrictions on the parameter space imposed by the above null and alternative hypotheses, do not violate the causality of the underlying VAR(d) model, that is, we assume that the condition $\det(\mathcal{A}(z)) \neq 0$ for all $z \leq 1$, is satisfied under H_0 and under H_1 .

In order to test the above hypotheses and to avoid problems associated with inverting large scale covariance matrices, we propose to use the max-type test statistic

$$T_n = \max_{(j,r,s) \in G} \left\{ \frac{\sqrt{n}|\hat{A}_{j;r}^{(s, \text{de})}|}{\widehat{s.e.}(j, r, s)} \right\}. \quad (18)$$

For $\alpha \in (0, 1)$ small, let $m_{n,\alpha}$ be the upper α -quantile of the distribution of T_n under the assumption that the null hypothesis is true. H_0 is then rejected if $T_n > m_{n,\alpha}$. Critical values of the test T_n can be obtained using the model-based bootstrap procedure proposed in this paper. For this, the estimated and sparsified VAR model used to generate the pseudo time series $X_1^*, X_2^*, \dots, X_n^*$ is modified in such a way that it satisfies the null hypothesis. This is important for a good size and power behavior of the bootstrap based test. To achieve this goal, the parameter matrices $A^{(s)}$, $s = 1, 2, \dots, d$, are estimated under the restrictions imposed by the null hypothesis, that is under the constraints $\hat{A}_{j;r}^{(s)} = 0$ for all indices $(j, r, s) \in G$. Using these restrictions on the matrices $\hat{A}^{(s)}$, the bootstrap algorithm proposed in Section 3.1 is then applied to generate pseudo time series $X_1^*, X_2^*, \dots, X_n^*$ from which the bootstrap analogue of T_n under validity of H_0 is calculated and which is given by

$$T_n^* = \max_{(j,r,s) \in G} \left\{ \frac{\sqrt{n}|\hat{A}_{j;r}^{*(s, \text{de})}|}{\widehat{s.e.}^*(j, r, s)} \right\}. \quad (19)$$

Let $m_{n,\alpha}^*$ be the upper α -quantile of the distribution of T_n^* . The bootstrap based test proceeds then by rejecting H_0 if $T_n > m_{n,\alpha}^*$. As the following theorem shows, the bootstrap succeeds in consistently estimating the distribution of T_n under the null, justifying, therefore, the use of the bootstrap critical values $m_{n,\alpha}^* = F_{T_n^*}^{-1}(1 - \alpha)$ for performing the test.

Theorem 6. *If Assumption 1 under validity of the null hypothesis is fulfilled then, as $n \rightarrow \infty$,*

$$\sup_{x \in \mathbb{R}} |P_{H_0}(T_n \leq x) - P(T_n^* \leq x | X_1, X_2, \dots, X_n)| = o_P(1), \quad (20)$$

where $P_{H_0}(T_n \leq \cdot)$ denotes the distribution function of T_n when H_0 is true.

We can extend the applicability of the bootstrap procedure proposed also to the case where the set G of null hypotheses considered increases to infinity, at an appropriate rate, as n increases to infinity. This is possible since a max-type test statistic is used such that Gaussian approximation results can be applied, see Zhang and Wu [45], Zhang et al. [46]. To state this dependence on the sample size, we write G_n for G . We then have the following theorem.

Theorem 7. *Let $|G_n| = O(n^b)$, $b > 0$ such that $n^{1/q-1/2+b/q}((b+1)\log(n))^{3/2} = o(1)$. Then under the assumptions of Theorem 6 we have that, as $n \rightarrow \infty$,*

$$\sup_{c \in \mathbb{R}} \left| P^* \left(\max_{(j,r,s) \in G_n} \left| \frac{\sqrt{n} |\hat{A}_{j;r}^{*(s,de)}|}{\widehat{s.e.}^*(j,r,s)} \right| \leq c \right) - P_{H_0} \left(\max_{(j,r,s) \in G_n} \left| \frac{\sqrt{n} |\hat{A}_{j;r}^{(s,de)}|}{\widehat{s.e.}(j,r,s)} \right| \leq c \right) \right| = o_P(1).$$

4. Numerical results

We investigate the finite sample performance of the bootstrap procedure proposed to infer properties of the sparse VAR model by means of simulations and we also discuss a real-life data application. Notice that we can use our bootstrap procedure to construct confidence intervals for the coefficients of the VAR model as well as to perform tests of statistical hypotheses about the VAR parameters. In this section, we focus on the problem of testing hypotheses about groups of model parameters.

All results presented in this section are based on implementations in *R* (R Core Team [33]) of the procedures proposed in this paper. In the simulations as well as in the real-life data example, the estimator $\hat{A}_{j;r}^{(s,rel)}$ of $A_{j;r}^{(s)}$ is based on the adaptive lasso; see Section 4 in Kock and Callot [23]. Furthermore, the estimator $\hat{A}_{j;r}^{(s)}$ used in our simulations, was obtained by thresholding the adaptive lasso estimator with tuning parameter λ_n and threshold parameter $a_n = \lambda_n$. The same estimator was used for the construction of the de-sparsified estimators in the bootstrap procedure. The Bayesian Information Criterion (BIC) has been used to select the tuning parameter λ_n . More specifically, the adaptive lasso implementation with BIC selection of λ_n of the *HDeconometrics* package (Garcia et al. [19]) which uses the *glmnet* package (Simon et al. [35]) has been applied. Finally, the covariance matrix Σ_ε of the innovations has been estimated using (9), where the threshold parameter b_n has been chosen by cross-validation and using the implementation of the package *FinCovRegularization* (Yan and Lin [43]).

In our numerical investigations, we have experienced that in situations where the set of null hypotheses is considerably large, that is, $|G|$ is of the same order as n , the bootstrap-based test and despite its consistency, seems to be somehow conservative. This is due to a finite sample bias appearing in estimating the upper percentage point of the distribution of T_n under H_0 . To improve the finite sample behavior the bootstrap based test, and inspired by Efron [15] for the construction of confidence intervals, see also Efron and Tibshirani [16], Section 14.3, we adapt to the testing context a bias correction procedure. This procedure is described in more detail in Supplementary Material B of the Supplemental Material and leads to the following bias-corrected bootstrap percentage point $m_{n,\alpha}^{*(B)} = F_{T_n^*}^{-1}(\Phi(\sqrt{2} \cdot \Phi^{-1}(P(T_n^+ \leq T_n^*)) + z_\alpha))$, where Φ is the distribution function of the standard normal and $\Phi(z_\alpha) = 1 - \alpha$. Notice that estimation of the quantity $P(T_n^+ \leq T_n^*)$ used in this bias correction procedure requires the implementation of a second bootstrap experiment; see Supplementary Material B for more details.

4.1. Simulations

Example 1. We generated time series $X_1, X_2, \dots, X_n$ from the VAR(1) model, $X_t = AX_{t-1} + \varepsilon_t$, with i.i.d. $\varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$, where the coefficient matrices A and Σ_ε possess a cluster (or block) structure. That is, only the coefficients within a small number of clusters are allowed to be different from zero; see also Han et al. [20] for the use of such VAR models. Each block is of size 20×20 and is given by the matrices A_{BLOCK}^ξ and Σ_{BLOCK} in Supplementary Material D. Four entries on the main diagonal of A_{BLOCK}^ξ are specified by the choice of the parameter ξ . This parameter controls in some sense the level of dependence of the generated VAR model obtained. Two values $\xi = 0.6$ and $\xi = 0.9$ are considered, which lead to the maximal absolute eigenvalues $\lambda_{\max} = 0.7$ and $\lambda_{\max} = 0.9$, respectively, of the coefficient matrix A . Note that the information that A possesses a cluster structure was not used in the subsequent inference procedure. Therefore, the same results could be obtained for the case where the indices are shuffled randomly; see Figure 2 in Supplementary Material D for an illustration of the matrix A and of a randomly shuffled version. The dimension p of the VAR process is set equal to 20, 100 and 200 where the case $p = 20$ results in a single cluster.

Let $G_1 = \{(i, j) : i \in \{1, \dots, 20\}, j \in \{p - 20 + 1, \dots, p\}\}$ be the first set and $G_2 = \{(i, j) : i \in \{1, \dots, 10\}, j \in \{p - 10 + 1, \dots, p\}\}$ be a second set of indices which correspond to the upper-right corner of the matrix A . Note that $|G_1| = 400$ while $|G_2| = 100$. We consider for $k \in \{1, 2\}$, the testing problem,

$H_0: A_{j:r} = 0$ for all $(j, r) \in G_k$, against

$H_1: \text{There exists } (j, r) \in G_k \text{ such that } A_{j:r} \neq 0.$

For $k = 1$, H_0 corresponds to the case that the time series belonging to the first cluster, that is, $X_{t,1}, \dots, X_{t,20}$, are not directly influenced by the set of lagged time series $X_{t-1,p-20+1}, \dots, X_{t,p}$ belonging to the last cluster. A similar interpretation occurs for $k = 2$ in which H_0 corresponds to the case where the first ten time series are not directly influenced by the last ten lagged time series. For $p = 20$ only the case $k = 2$ is considered since for $k = 1$ the corresponding null hypothesis is that X_t is a white noise process. We investigate the performance of the bootstrap based test under the null as well as its power against various alternatives. The alternatives considered refer to the case where only one coefficient in the set G is set different to zero and equal to $\delta_a = 0.3$. Notice that $\delta_a = 0$ corresponds to the null hypothesis. The results obtained for 500 repetitions and $B = 1000$ bootstrap replications are presented in Table 1 (size) and Table 2 (power).

As Table 1 shows, the bootstrap-based test without the proposed bias correction, seems to be conservative in most of the cases and the differences between the empirical and the nominal level increase as the size of the set G increases. The degree of persistence as well as the dimension of the process seem not to affect the size behavior of the test. However, using the bias correction, the behavior of the bootstrap-based test improves considerably for all sizes of the set G considered. Table 2 presents the empirical power of the bootstrap based test. As it can be seen, in all settings considered, the bias correction improves the power behavior of the test, where (as expected) the test is more powerful for the set G_2 than for the more larger set G_1 . Notice that for the sample size of $n = 128$ observations, detecting the deviation from the null which is due to the fact that only one out of the $|G_1| = 400$ elements is set equal to 0.3, is a very challenging inference problem. However, even in this case, the test has power and its power improves considerably as the sample size increases. Finally, as in the case of the size, the dimension of the VAR process seems to affect only slightly the power of the test.

Table 1. Rejection frequencies under H_0 ($\delta_a = 0.0$), for time series stemming from the VAR model of Example 1 with dimension $p = 20, 100, 200$ and sample sizes of $n = 128$ and 200 observations

		G_2						G_1			
p		20		100		200		100		200	
α		0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10
n	λ	With bias correction									
128	0.7	0.04	0.08	0.05	0.11	0.03	0.07	0.05	0.11	0.03	0.06
	0.9	0.04	0.10	0.05	0.10	0.05	0.10	0.04	0.10	0.05	0.09
200	0.7	0.04	0.10	0.05	0.11	0.06	0.09	0.04	0.10	0.06	0.11
	0.9	0.06	0.12	0.05	0.10	0.05	0.12	0.06	0.12	0.04	0.09
n	λ	Without bias correction									
128	0.7	0.03	0.06	0.03	0.08	0.02	0.05	0.03	0.06	0.01	0.04
	0.9	0.04	0.07	0.04	0.08	0.03	0.07	0.02	0.05	0.03	0.06
200	0.7	0.04	0.08	0.04	0.10	0.05	0.08	0.02	0.07	0.04	0.08
	0.9	0.05	0.10	0.03	0.07	0.04	0.09	0.04	0.09	0.02	0.06

4.2. A real-life data example

In Farmer [18], the question has been discussed whether the stock market affects the labor market. In the aforementioned paper, the stock market is represented by the S&P 500 index and the labor market by the unemployment rate, that is, a bivariate time series is used to investigate the question of interest. However, restricting the analysis to a bivariate system might be problematic since it implies a loss

Table 2. Rejection frequencies under H_1 ($\delta_a = 0.3$), for time series stemming from the VAR model of Example 1 with dimension $p = 20, 100, 200$ and sample sizes of $n = 128$ and 200 observations

		G_2						G_1			
p		20		100		200		100		200	
α		0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10
n	λ	With bias correction									
128	0.7	0.61	0.75	0.64	0.77	0.62	0.79	0.27	0.43	0.28	0.43
	0.9	0.61	0.74	0.62	0.79	0.61	0.78	0.25	0.41	0.18	0.32
200	0.7	0.95	0.99	0.95	0.97	0.96	0.98	0.83	0.90	0.82	0.88
	0.9	0.95	0.99	0.96	0.98	0.96	0.98	0.82	0.89	0.76	0.88
n	λ	Without bias correction									
128	0.7	0.51	0.68	0.52	0.68	0.50	0.68	0.15	0.25	0.12	0.24
	0.9	0.51	0.68	0.52	0.70	0.52	0.69	0.12	0.24	0.08	0.17
200	0.7	0.93	0.97	0.94	0.97	0.94	0.96	0.73	0.85	0.73	0.82
	0.9	0.94	0.98	0.95	0.98	0.94	0.97	0.74	0.84	0.66	0.81

of information due to the fact that a variety of time series exist which describe the activities in the two different macroeconomic sectors. The corresponding dependence structures might be much more complicated than those captured by the particular bivariate time series considered. Therefore, and in order to get a more detailed and deeper inside into the relations between the labor and the stock market, we consider the Federal Reserve Economic Data (FRED) of St. Louis Fed's main, which is publicly available at <http://research.stlouisfed.org/fred2/>. This data set contains 31 time series describing the activities in the labor market and 5 time series describing the stock market. Furthermore, to take into account the fact that other macroeconomic variables may also exist which simultaneously influence the labor and the stock market, all available time series in the aforementioned data set are considered, which refer to a wide range of different economic activities. The entire data set considered contains $p = 124$ time series and a complete description of all time series used is given in Supplementary Material C of this paper. The techniques proposed in McCracken and Ng [29] have been adopted to transform this set of time series to stationary. Furthermore, to ensure comparability of the data used in our analysis with the data set used by Farmer [18], the monthly observations have been aggregated to quarterly data where the time period considered begins by the fourth quarter of 1979 and ends by the first quarter of 2011. The number of available observations is then equal to $n = 126$. As already mentioned, the question of interest is whether the financial sector, that is the stock market, influences the labor market. Following Farmer [18], a vector autoregressive model of order 1 has been used and the following hypotheses have been considered:

$$\begin{aligned} H_0: & A_{j;r} = 0 \text{ for all } (j, r) \in G, \\ H_1: & \text{There exists } (j, r) \in G \text{ such that } A_{j;r} \neq 0, \end{aligned}$$

where G is the set containing the indices corresponding to the two economic sectors, that is $G = \{(j, r) : j \in \text{LABOR and } r \in \text{STOCK}\}$. Here *STOCK* denotes the set of indices referring to the time series of the stock market and *LABOR* to the time series of the labor market. Notice that $|G| = 155$. To test the above pair of hypotheses, the test statistic described in Section 3.2 with $B = 2000$ bootstrap replications and bias corrected percentage points have been used.

The degree of sparsity obtained depends on the choice of the regularization parameters. As mentioned in the previous section, we use the adaptive lasso with regularization parameter λ_n and threshold parameter $a_n = \lambda_n$ to estimate the coefficient matrix, while the covariance of the innovations Σ_ε is estimated as in (9), with the threshold parameter b_n chosen by cross-validation. For instance, for $\lambda_n = 0.25$ we obtain a sparsity in the coefficient matrix A of 0.56%, for $\lambda_n = 0.1$ the sparsity obtained is 2.7% and if λ_n is chosen by BIC, it leads to a sparsity level of 1.43%. The p -values of the test seem not to be largely affected by the choice of this regularization parameter. This is demonstrated in

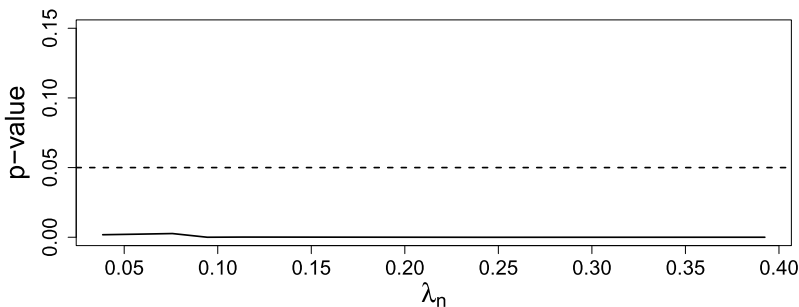


Figure 1. P -values (vertical axis) of the test T_n described in Section 3.2 applied to the FRED data set for several values of the regularization parameter λ_n (horizontal axis).

Figure 1 where the p -values of the bootstrap based test proposed, are shown for different values of the regularization parameter λ_n .

As it is seen from this figure, the null hypothesis of interest is rejected at the commonly used $\alpha = 0.05$ level, for all values of λ_n considered. If λ_n is chosen by BIC, the test gives a p -values of less than 0.0002. Furthermore, our bootstrap procedure identifies the following coefficients $A_{j;r}$, for $j \in \text{LABOR}$ and $r \in \text{STOCK}$, as different from zero: $A_{\text{USGOOD},S.P.500} = 0.13$, $A_{\text{MANEMP},S.P.500} = 0.18$, $A_{\text{CLAIMSX},S.P.\text{div.yield}} = 0.32$, and $A_{\text{NDMANEMP},S.P.\text{indust.}} = 0.16$. Thus, our analysis leads to the conclusion that the hypothesis that the stock market does not influence the labor market should be rejected. Notice that this conclusion is also supported by the findings reported in Phelps [32] and Farmer [18].

5. Conclusions

In this paper, high-dimensional and sparse vector autoregressive models have been considered. We have first adopted the concept of de-biasing to the high-dimensional VAR(d) time series context and have considered de-biased respectively de-sparsified estimators of the autoregressive parameters. The asymptotic distribution of the de-sparsified estimators has been derived under general conditions. Furthermore, a bootstrap procedure has been proposed which is asymptotically able to generate pseudo time series that appropriately imitate the dependence structure and the sparsity properties of the underlying high-dimensional VAR(d) process. Asymptotic validity of the bootstrap procedure proposed for estimating the distribution of de-sparsified estimators has been established. Furthermore, an appropriately modified version of the bootstrap procedure has been used for testing hypotheses about groups of model parameters. We allow for these groups of model parameters to increase to infinity with sample size. Validity of the bootstrap procedure also has been established for this case. Finally, we have demonstrated by means of numerical investigations, the good finite sample behavior of the bootstrap-based inference procedure proposed and we have analyzed an interesting real-life data set.

6. Proofs

Proofs which are not given in this section are presented in Supplementary Material E of the Supplementary Material.

Lemma 8. *Let $\mathbb{E} = (I_p, 0, \dots, 0)^\top \in \mathbb{R}^{dp \times p}$. If Assumption 1(i) to (iv) holds true, then the estimator of $\Gamma(h)$ given by*

$$\widehat{\Gamma}^{(st)}(h) = \sum_{j=0}^{\infty} \widehat{\mathbb{A}}^{j+h} \mathbb{E} \widehat{\Sigma}_\varepsilon \mathbb{E}^\top (\widehat{\mathbb{A}}^\top)^j = \widehat{\mathbb{A}}^h \text{vec}^{-1}((I_{(dp)^2} - \widehat{\mathbb{A}} \otimes \widehat{\mathbb{A}})^{-1} \text{vec}(\mathbb{E} \widehat{\Sigma}_\varepsilon)), \quad h \geq 0, \quad (21)$$

and $\widehat{\Gamma}^{(st)}(h) = (\widehat{\Gamma}^{(st)}(-h))^\top$ for $h < 0$ satisfies $\|\widehat{\Gamma}^{(st)}(h) - \Gamma^{(st)}(h)\|_\infty = O_P(k_2(p)k_1(p)^4 \times \sqrt{g(p, d, n)/n})$. Furthermore, if Assumption 1(v) holds true, we also have $\|(\widehat{\Gamma}^{(st)}(h))^{-1} - (\Gamma^{(st)}(h))^{-1}\|_\infty = O_P(k_2(p)k_1(p)^4 k_3(p)^2 \sqrt{g(p, d, n)/n})$.

Proof of Lemma 8. In order to simplify notation, we set $h = 0$. Note that for $j \in \mathbb{N}$ we have $\widehat{\mathbb{A}}^j - \mathbb{A}^j = (\widehat{\mathbb{A}} - \mathbb{A} + \mathbb{A})(\widehat{\mathbb{A}}^{j-1} - \mathbb{A}^{j-1}) + (\mathbb{A} - \widehat{\mathbb{A}})\mathbb{A}^{j-1}$. Using this recursive formula, we obtain $\widehat{\mathbb{A}}^j - \mathbb{A}^j = \sum_{s=0}^{j-1} [(\widehat{\mathbb{A}} - \mathbb{A}) + \mathbb{A}]^s (\widehat{\mathbb{A}} - \mathbb{A}) \mathbb{A}^{j-1-s}$. Since $\|\mathbb{A} - \widehat{\mathbb{A}}\|_\infty = O_P(k_1(p)^{1.5} \sqrt{g(p, d, n)/n})$ by Assumption 1(iii), we have $\|\widehat{\mathbb{A}}^j - \mathbb{A}^j\|_\infty = \|\sum_{s=0}^{j-1} [(\widehat{\mathbb{A}} - \mathbb{A}) + \mathbb{A}]^s (\widehat{\mathbb{A}} - \mathbb{A}) \mathbb{A}^{j-1-s}\|_\infty =$

$O_P(j\lambda^{j-1}k_1(p)^{3.5}\sqrt{g(p,d,n)/n})$. We have by the same arguments $\|\mathbb{A}^j - \widehat{\mathbb{A}}^j\|_1 = \|(\mathbb{A}^\top)^j - (\widehat{\mathbb{A}}^\top)^j\|_\infty = O_P(j\lambda^{j-1}k_1(p)^{3.5}\sqrt{g(p,d,n)/n})$. Furthermore, note that Assumption 1(ii) implies $\sum_{j=0}^\infty \|\mathbb{A}^j\|_g = O(k_1(p))$ and also $\sum_{j=0}^\infty \|\widehat{\mathbb{A}}^j - \mathbb{A}^j\|_g = O(k_1(p)^{3.5}\sqrt{g(p,d,n)/n})$ for $g \in \{1, \infty\}$. We then have $\|\Gamma^{(st)}(0) - \widehat{\Gamma}^{(st)}(0)\|_\infty \leq \|\sum_{j=0}^\infty (\mathbb{A}^j - \widehat{\mathbb{A}}^j)\mathbb{E}\Sigma_\varepsilon\mathbb{E}^\top(\mathbb{A}^\top)^j\|_\infty + \|\sum_{j=0}^\infty (\widehat{\mathbb{A}}^j)\mathbb{E}(\Sigma_\varepsilon - \widehat{\Sigma}_\varepsilon)\mathbb{E}^\top(\mathbb{A}^\top)^j\|_\infty + \|\sum_{j=0}^\infty (\widehat{\mathbb{A}}^j)\mathbb{E}\widehat{\Sigma}_\varepsilon\mathbb{E}^\top((\mathbb{A}^\top)^j - (\widehat{\mathbb{A}}^\top)^j)\|_\infty = O_P(k_1(p)^{4.5}k_2(p)\sqrt{g(p,d,n)/n})$.

Using $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$, we have $\|(\Gamma^{(st)}(0))^{-1} - (\widehat{\Gamma}^{(st)}(0))^{-1}\|_\infty \leq \|(\Gamma^{(st)}(0)) - \widehat{\Gamma}^{(st)}(0)\|_\infty \|(\Gamma^{(st)}(0))^{-1} - (\widehat{\Gamma}^{(st)}(0))^{-1}\|_\infty + \|(\Gamma^{(st)}(0))^{-1}\|_\infty \|\Gamma^{-1}\|_\infty$. For n large enough such that $\|(\Gamma^{(st)}(0)) - \widehat{\Gamma}^{(st)}(0)\|_\infty < \|(\Gamma^{(st)}(0))^{-1}\|_\infty^{-1}$, we have $\|(\Gamma^{(st)}(0))^{-1} - (\widehat{\Gamma}^{(st)}(0))^{-1}\|_\infty \leq \|(\Gamma^{(st)}(0)) - \widehat{\Gamma}^{(st)}(0)\|_\infty \|(\Gamma^{(st)}(0))^{-1}\|_\infty^2 (1 - \|(\Gamma^{(st)}(0))^{-1}\|_\infty \|(\Gamma^{(st)}(0)) - \widehat{\Gamma}^{(st)}(0)\|_\infty)^{-1}$. $\square$

Lemma 9. Let $\widehat{A}^{(s)}$, $s = 1, \dots, d$ be the estimator of $A^{(s)}$, $s = 1, \dots, d$ given by

$$\widehat{A}_{j;r}^{(s)} = \widehat{A}_{j;r}^{(s,\text{re})} \mathbb{1}(|\widehat{A}_{j;r}^{(s,\text{re})}| \geq a_n), \quad j, r = 1, \dots, p, s = 1, \dots, d.$$

If $P(\max_{i,j,s} |e_i^\top (\widehat{A}^{(s,\text{re})} - A)e_j| > r_n) = o(1)$, and $a_n = Cr_n$ for some constant $C > 0$, then $\sum_{s=1}^d \|\widehat{A}^{(s)} - A\|_\infty = O_P(k_1(p)r_n)$, $\sum_{s=1}^d \|\widehat{A}^{(s)} - A\|_1 = O_P(k_1(p)r_n)$, $\max_j \sum_{s=1}^d \|\widehat{A}^{(s)}e_j\|_0 = O_P(k_1(p))$ and $\max_j \sum_{s=1}^d \|e_j^\top \widehat{A}^{(s)}\|_0 = O_P(k_1(p))$.

Proof. The proof uses ideas similar to those used in the proof of Theorem 1 in Bickel and Levina [3] and Theorem 1 in Cai and Liu [5]. Let $J_{i,s} = \{j = 1, \dots, p : A_{i,j}^{(s)} \neq 0\}$ be the set of indices for which the entries of the i th row of $A^{(s)}$ are non-zero. In the event of $\max_{i,j,s} |e_i^\top (\widehat{A}^{(s,\text{re})} - A)e_j| \leq r_n$, we have $\sum_{s=1}^d \|\widehat{A}^{(s)} - A^{(s)}\|_\infty \leq \sum_{s=1}^d \max_i \sum_{j \in J_{i,s}} |\widehat{A}_{i;j}^{(s,\text{re})}| \mathbb{1}\{|\widehat{A}_{i;j}^{(s,\text{re})}| \geq a_n\} + \sum_{s=1}^d \max_i \sum_{j \in J_{i,s}^c} |\widehat{A}_{i;j}^{(s,\text{re})} - A_{i;j}| \times \mathbb{1}\{|\widehat{A}_{i;j}^{(s,\text{re})}| \geq a_n\} + \sum_{s=1}^d \max_i \sum_{j \in J_{i,s}} |A_{i;j}^{(s)}| \mathbb{1}\{(|\widehat{A}_{i;j}^{(s,\text{re})}| < a_n, |A_{i;j}^{(s)}| \neq 0)\} \leq k_1(p)r_n$. Since this event occurs with probability converging to one, we obtain $\sum_{s=1}^d \|\widehat{A}^{(s)} - A^{(s)}\|_\infty = O_P(k_1(p)r_n)$. In the same way, we obtain $\sum_{s=1}^d \|\widehat{A}^{(s)} - A^{(s)}\|_1 = O_P(k_1(p)r_n)$.

Furthermore, we have $\sum_{s=1}^d \max_i \sum_{j=1}^p \mathbb{1}\{|\widehat{A}_{i;j}^{(s)}| \geq a_n\} \leq \sum_{s=1}^d \max_i \sum_{j \in J_{i,s}} \mathbb{1}\{|\widehat{A}_{i;j}^{(s,\text{re})}| \geq a_n\} + \sum_{j \in J_{i,s}^c} \mathbb{1}\{|\widehat{A}_{i;j}^{(s,\text{re})}| \geq a_n\} \leq k_1(p) + \sum_{s=1}^d \max_i \sum_{j \in J_{i,s}^c} |\widehat{A}_{i;j}^{(s,\text{re})} - A_{i;j}|/a_n \leq k_1(p) + \sum_{s=1}^d \|\widehat{A}^{(s,\text{re})} - A^{(s)}\|_\infty/a_n = O_P(k_1(p))$. Similarly, we have $\sum_{s=1}^d \max_j \sum_{i=1}^p \mathbb{1}\{|\widehat{A}_{i;j}^{(s)}| \geq a_n\} = O_P(k_1(p))$. Thus, $\widehat{A}^{(s)}$ gives a row-wise and column-wise sparse estimator of $A^{(s)}$ and the assertion follows. $\square$

Lemma 10. Let $\widehat{\varepsilon}_t = \widetilde{\varepsilon}_t - 1/(n-d) \sum_{t=d}^n \widetilde{\varepsilon}_t$, $\widetilde{\varepsilon}_t = X_t - \sum_{s=1}^d \widehat{A}^{(s)} X_{t-j} = X_t - \Xi W_{t-1}$, $t = d+1, \dots, n$, be the centered estimated residuals of $\varepsilon_1 \sim (0, \Sigma)$, $\max_j \|\Sigma e_j\|_0 = k_2(p)$ and $\|\Sigma\|_1 = O(k_2(p))$. Recall the definitions of $\widehat{\Sigma}_\varepsilon$ and $\widetilde{\Sigma}_\varepsilon$ given in (9). If Assumption 1(i) and 1(iii) are satisfied, then

$$\max_{i,j} |e_i^\top (\widetilde{\Sigma}_\varepsilon - \Sigma_{\varepsilon,n})e_j| = O_P(\sqrt{g(p,d,n)/n}k_1(p)^2(\sqrt{g(p,d,n)/n} + \sqrt{\log(p)/n})), \quad (22)$$

where $\Sigma_{\varepsilon,n} = n^{-1} \sum_{t=1}^n \varepsilon_t \varepsilon_t^\top$.

If the ε_t 's are Gaussian and $b_n = C(\sqrt{\log(p)/n})$ for some constant $C > 0$, then

$$\|\widehat{\Sigma}_\varepsilon - \Sigma_\varepsilon\|_1 = O_P(k_2(p)\sqrt{\log(p)/n} + k_1(p)^2 \log(p)/n) \quad (23)$$

and $\max_{1 \leq j \leq p} \|\widehat{\Sigma}_\varepsilon e_j\|_0 = O_P(k_2(p))$.

The main result of Lemma 10 is (22) since it states that the error of using estimated residuals instead of the true ones is of higher order. With a more refined thresholding approach, see, for example, Cai and Liu [5], (23) can be extended to non-Gaussian innovations.

Lemma 11. Let $Y_{t;i} = G_i(\varepsilon_t, \varepsilon_{t-1}, \dots), i = 1, \dots, \tilde{p}, t \in \mathbb{Z}$, be some process generated causally by the i.i.d. processes $\{\varepsilon_t\}$ for some function $G = (G_1, \dots, G_{\tilde{p}})$. Furthermore, denote by $Y_{t;i}^{(k)} = G_i(\varepsilon_t, \varepsilon_{t-1}, \dots, \varepsilon_{t-k+1}, \varepsilon'_{t-k}, \varepsilon_{t-k-1}, \varepsilon_{t-k-2}, \dots)$ the process where ε_{t-k} is replaced by an i.i.d. copy of it. Furthermore, define the physical dependence coefficients, see Wu [41], Zhang and Wu [45], Liu et al. [27], in the following way. Let $\delta_{k,q,i} = \|Y_{0;i} - Y_{0;i}^{(k)}\|_{E,q}, k \geq 0, \Delta_{m,q,i} = \sum_{k=m}^{\infty} \delta_{k,q,i}, \|Y_{t;i}\|_{q,\alpha} = \sup_{m \geq 0} (m+1)^\alpha \Delta_{m,q,i}$, $\Psi_{q,\alpha} = \max_{1 \leq i \leq \tilde{p}} \|Y_{t;i}\|_{q,\alpha}$, $\Upsilon_{q,\alpha} = (\sum_{i=1}^{\tilde{p}} (\sup_{m \geq 0} (m+1)^\alpha \Delta_{m,q,i})^q)^{1/q}$, $\omega_{k,q} = \|\max_{1 \leq i \leq \tilde{p}} \|Y_{t;i} - Y_{t;i}^{(k)}\|_q\|_q$, $\Omega_{m,q} = \sum_{k=m}^{\infty} \omega_{k,q}$, and $\|Y\|_{q,\alpha} = \sup_{m \geq 0} (m+1)^\alpha \Omega_{m,q}$. Furthermore, let $v_q = \sum_{j=1}^{\infty} (jq^{2-1} \omega_{k,q})^{1/(q+1)}$.

If Assumption 1(i),(ii),(vi) is satisfied, then the following assertions hold true:

1. For the process $\{(e_r^\top (\Gamma^{(st)}(0))^{-1} W_t W_t^\top e_r - 1)_{r=1, \dots, dp}, t \in \mathbb{Z}\}$ we have $v_{q/2} < \infty$.
2. Let $G_n = \{(j_1, r_1, s_1), \dots, (j_{n_g}, r_{n_g}, s_{n_g})\}$ be some set of indices and let further $Y_{t,i} := e_{(s_i-1)p+r_i} (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1;j_i} / s.e.(r_i, j_i, s_i), i = 1, \dots, n_g$. Then for some $\alpha > 1/2 - 1/q$ we have $\Psi_{q,\alpha} < \infty, \Upsilon_{q,\alpha} = O(n_g^{1/q}), \|Y\|_{q,\alpha} = O(k_3(p)k_1(p)), v_q < \infty$.

Lemma 12. Let $W_t = (X_t^\top, \dots, X_{t-d+1}^\top)^\top = \sum_{j=0}^{\infty} \mathbb{A}^j \mathbb{U}_{t-j}$ and suppose that Assumption 1(i),(ii),(v), (vi) is true. Then, for some vector $v \in \mathbb{R}^{dp}, \|v\|_1 = 1$ we have

$$\max_{r=1, \dots, dp} \left| \frac{1}{n} e_r^\top \left[\sum_{t=d}^{n-1} (\Gamma^{(st)}(0))^{-1} W_t W_t^\top - I_{dp} \right] v \right| = O_P(\sqrt{\log(p)/n}).$$

Lemma 13. Let $\{Y_t, t \in \mathbb{Z}\}$ be a centered, p -dimensional stationary process and let $v < \infty$ as given in Lemma 11. If $p/((\log p)^{q/2} n^{q/2-1}) = O(1)$, then

$$\max_{1 \leq j \leq p} \left| e_j^\top \frac{1}{\sqrt{n}} \sum_{t=1}^n Y_t \right| = O_P(\sqrt{\log p}).$$

Lemma 14. Let $G_n = \{(j_1, r_1, s_1), \dots, (j_{n_g}, r_{n_g}, s_{n_g})\}$ be some set of indices and let $Y_{t;(r_i, j_i, s_i)} = e_{(s_i-1)p+r_i} (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1;j_i} / s.e.(r_i, j_i, s_i), i = 1, \dots, n_g$. Set $h_1(x) = x, h_2(x) = -x$ and $h_3(x) = |x|$. If Assumption 1(i)–(vi) hold true, then

$$\max_{(j,r,s) \in G_n} \frac{\sqrt{nh_i}(\hat{A}_{j;r}^{(s,de)} - A_{j;r}^{(s)})}{\widehat{s.e.}(j, r, s)} = \max_{(j,r,s) \in G_n} h_i \left(\frac{1}{\sqrt{n}} \sum_{t=d}^{n-1} Y_{t;(r,j,s)} \right) + o_P(1), \quad i = 1, 2, 3.$$

Proof. We proof the assertion for h_3 , the other cases can be handled analogously. Denote by $\mathcal{I} = \{1, \dots, n_g\}$ a set of indices corresponding to G_n and let $\tilde{r} = (s-1)p + r$ for $(j, r, s) \in G_n$. We have $(\hat{A}_{j;r}^{(s,de)} - A_{j;r}^{(s)}) = \sqrt{n}(\sum_{t=d}^{n-1} \hat{Z}_{t;\tilde{r}} W_t W_t^\top)^{-1} (\sum_{t=d}^{n-1} \hat{Z}_{t;\tilde{r}} \varepsilon_{t+1;j}) + \sqrt{n}(\sum_{t=d}^{n-1} \hat{Z}_{t;\tilde{r}} W_t W_t^\top)^{-1} (\sum_{t=d}^{n-1} \hat{Z}_{t;\tilde{r}} \times W_{t;-\tilde{r}}) (\Xi_{j;-\tilde{r}} - \hat{\Xi}_{j;-\tilde{r}}^{(re)})$.

Consider first that $\hat{Z}_{t;\tilde{r}} = e_{\tilde{r}}^\top (\hat{\Gamma}^{(st)}(0))^{-1} W_t / [e_{\tilde{r}}^\top (\hat{\Gamma}^{(st)}(0))^{-1} e_{\tilde{r}}] =: \omega_{\tilde{r}} e_{\tilde{r}}^\top (\hat{\Gamma}^{(st)}(0))^{-1} W_t$. The positive factor $\omega_{\tilde{r}}$ occurs in the numerator and in the denominator and can be omitted. Let $DN = 1/n(\sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\hat{\Gamma}^{(st)}(0))^{-1} W_t W_t^\top)$. Furthermore, observe that $DN = 1/n \sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\hat{\Gamma}^{(st)}(0))^{-1} W_t \times$

$W_t^\top e_{\tilde{r}} = 1/n(\sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\Gamma^{(st)}(0))^{-1} W_t W_t^\top e_{\tilde{r}} + \sum_{t=d}^{n-1} e_{\tilde{r}}^\top ((\widehat{\Gamma}^{(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}) W_t W_t^\top e_{\tilde{r}}) = 1 + O_P(k_2(p)k_1(p)^{4.5}k_3(p)^2\sqrt{g(p, d, n)/n})$, due to Lemma 12 and Lemma 8. Note further that

$$\begin{aligned}
 s.e.(j, r, s)^2 &= \Sigma_{\varepsilon, j; j} (e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} e_{(s-1)p+r}) \quad \text{and} \\
 \widehat{s.e.}(j, r, s)^2 &= \widehat{\Sigma}_{\varepsilon, j; j} (e_{(s-1)p+r}^\top (\widehat{\Gamma}^{(st)}(0))^{-1} e_{(s-1)p+r}).
 \end{aligned}$$

Thus, $\max_{(j, r, s) \in G_n} \widehat{s.e.}(j, r, s)^2 / s.e.(j, r, s)^2 = 1 + O_P(k_2(p)\sqrt{g(p, d, n)/n} + k_2(p)k_1(p)^{4.5}k_3(p)^2 \times \sqrt{g(p, d, n)/n})$ by Assumption 1(iv) and Lemma 8.

Furthermore, we have

$$\begin{aligned}
 &\max_{(j, r, s) \in G} \left| \frac{1}{\sqrt{n}} \left(\sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\widehat{\Gamma}^{(st)}(0))^{-1} W_t W_{t; -(\tilde{r})} (\Xi_{j; -(\tilde{r})} - \widehat{\Xi}_{j; -(\tilde{r})}^{(\text{re})}) \right) \right| \\
 &\leq \max_{(j, r, s) \in G} \left| \frac{1}{\sqrt{n}} \left(\sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\Gamma^{(st)}(0))^{-1} W_t W_{t; -(\tilde{r})} (\Xi_{j; -(\tilde{r})} - \widehat{\Xi}_{j; -(\tilde{r})}^{(\text{re})}) \right) \right| \\
 &\quad + \max_{(j, r, s) \in G} \left| \frac{1}{\sqrt{n}} \left(\sum_{t=d}^{n-1} e_{\tilde{r}}^\top ((\widehat{\Gamma}^{(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}) W_t W_{t; -(\tilde{r})} (\Xi_{j; -(\tilde{r})} - \widehat{\Xi}_{j; -(\tilde{r})}^{(\text{re})}) \right) \right| \\
 &\leq \max_{(j, r, s) \in G} \max_{s \neq \tilde{r}} \left| \frac{1}{\sqrt{n}} \sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\Gamma^{(st)}(0))^{-1} W_t W_{t; s} \right| \|\Xi - \widehat{\Xi}^{(\text{re})}\|_\infty \\
 &\quad + \sqrt{n} \max_s \left| \frac{1}{n} \sum_{t=d}^{n-1} W_{t; s} W_{t; s} \right| \|(\widehat{\Gamma}^{(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}\|_\infty \|\Xi - \widehat{\Xi}^{(\text{re})}\|_\infty \\
 &= O_P(k_1(p)\sqrt{\log(p)g(p, d, n)/n} + k_1(p)k_2(p)k_1(p)^{4.5}k_3(p)^2g(p, d, n)/\sqrt{n}),
 \end{aligned}$$

by Lemma 12, 13 and 8. Furthermore, we have $1/\sqrt{n} \sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\widehat{\Gamma}^{(st)}(0))^{-1} W_t \varepsilon_{t+1; j} = 1/\sqrt{n} \times \sum_{t=d}^{n-1} e_{\tilde{r}}^\top (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1; j} + 1/\sqrt{n} \sum_{t=d}^{n-1} e_{\tilde{r}}^\top ((\widehat{\Gamma}^{(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}) W_t \varepsilon_{t+1; j}$. Applying Lemma 11 to $\{W_t \varepsilon_{t+1}\}$ leads to $v_q < \infty$. Lemma 13 implies

$$\begin{aligned}
 &\max_{(j, r, s) \in G_n} \left| \frac{1}{\sqrt{n}} \sum_{t=d}^{n-1} e_{\tilde{r}}^\top ((\widehat{\Gamma}^{(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}) W_t \varepsilon_{t+1; j} \right| \\
 &\leq \|(\widehat{\Gamma}^{(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}\|_\infty \max_{(j, r, s) \in G_n} \max_{s_1, s_2} \left| \frac{1}{\sqrt{n}} \sum_{t=d}^{n-1} e_{s_1} W_t \varepsilon_{t+1; s_2} \right| \\
 &= O_P(\sqrt{\log(n_g)k_2(p)k_1(p)^{4.5}k_3(p)^2g(p, d, n)/n}).
 \end{aligned}$$

This completes the calculations to show that the nuisance terms are asymptotically negligible and the assertion follows. $\square$

Proof of Theorem 1. Applying Lemma 14 gives us

$$\sqrt{n}(\widehat{A}_{j;r}^{(s,de)} - A_{j;r}^{(s)}) = 1/\sqrt{n} \sum_{t=d}^{n-1} e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1;j} + o_P(1) =: 1/\sqrt{n} \sum_{t=d}^{n-1} Y_t + o_P(1).$$

Furthermore, we have by Lemma 11 that $\{Y_t\}$ fulfills the conditions of physical dependence, hence $v_q < \infty$. Note further that due to the i.i.d. property of $\{\varepsilon_t\}$ we have that $\{Y_t\}$ is an uncorrelated sequence which gives $\text{Var}(1/\sqrt{n} \sum_{t=d}^{n-1} Y_t) = n/(n-d) \Sigma_{\varepsilon;jj} (e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} e_{(s-1)p+r}^\top)$. Furthermore, Assumption 1(vi) ensures the existence of fourth order moments. Thus, a Lyapounov condition can be verified for the sequence $\{n^{-1/2} \sum_{t=d+1}^n Y_t\}$, establishing $1/\sqrt{n} \sum_{t=d+1}^n Y_t \xrightarrow{D} \mathcal{N}(0, \Sigma_{\varepsilon;jj} (e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} e_{(s-1)p+r}^\top))$ via an extension of the central limit theorem for functional dependent random variables, Theorem 3 of Wu [42], to triangular arrays; see also Theorem 27.3 of Billingsley [4]. Thus, the assertion follows by Slutsky's lemma. $\square$

Proof of Lemma 2. Let $\tilde{r}_1 = (s_1 - 1)p + r_1$ and $\tilde{r}_2 = (s_2 - 1)p + r_2$. Note further that due to the i.i.d. property of $\{\varepsilon_t\}$ we have that $\{W_{t_1} \varepsilon_t\}$ is an uncorrelated sequence. Thus, we have $\text{Cov}(e_{\tilde{r}_1}^\top (\Gamma^{(st)}(0))^{-1} W_{t-1} \varepsilon_t; j_1, e_{\tilde{r}_2}^\top (\Gamma^{(st)}(0))^{-1} t - 1 \varepsilon_t; j_2) = (\Sigma_{\varepsilon;jj} e_{\tilde{r}_1}^\top (\Gamma^{(st)}(0))^{-1} e_{\tilde{r}_2})$, the assertion follows Lemma 14. $\square$

Proof of Theorem 3. Denote by $*$ the corresponding quantities in the bootstrap world, that is $X_t^*, W_t^* = (X_{t-1}^*, \dots, X_{t-d}^*)$ and $\varepsilon_t^*, t = 1, \dots, n$. We show that, as $n \rightarrow \infty$, $\sqrt{n}(\widehat{A}_{j;r}^{*(s,de)} - \widehat{A}_{j;r}^{(s)}) \xrightarrow{D} \mathcal{N}(0, s.e.(j, r, s)^2)$ in probability, from which the assertion follows by the triangular inequality. To establish the above weak convergence, similar ideas as those used in the proof of Theorem 1 can be applied.

Since $\widehat{A}^{(s)}, s = 1, \dots, d$ and $\widehat{\Sigma}_\varepsilon$ give an sparse estimate, see Lemma 9 and Lemma 10 we have that $\{X_t^*\}$ is a sparse VAR(d) process with Gaussian i.i.d. innovations. Furthermore, due to Assumption 1(ii) and (iii) this process is (asymptotically) causal and we have $X_t^* = (\widehat{A}^{*1}, \dots, \widehat{A}^{*d})(X_{t_1}^{*\top}, \dots, X_{t-d}^{*\top})^\top + \varepsilon_t^* =: \Xi^* W_t^* + \varepsilon_t^*$. Thus, Lemma 12, 13, and 11 hold for this processes as well. A regularized estimator $\widehat{\Xi}^{*(re)} = (\widehat{A}^{*(s,re)}, s = 1, \dots, d)$ similarly as in the construction of $\{\widehat{A}^{(de)}\}$ but based on the pseudo time series $X_1^*, \dots, X_n^*$ could be used to estimate $\Xi^* = (\widehat{A}^{(s)}, s = 1, \dots, d)$ in the bootstrap world. For $l \in \{1, \infty\}$, the Gaussianity of $\{\varepsilon_t^*\}$ gives $\|\widehat{\Xi}^{*(re)} - \Xi^*\|_l = O_P(k_1(p)^{1.5} \sqrt{\log(p)/n})$. Note that $\|\Xi^* - \Xi\|_l = O_P(k_1(p)^{3/2} \sqrt{g(p, d, n)/n})$. Hence, we have by triangular inequality and by Assumption 1(iii) $\|\widehat{\Xi}^{*(re)} - \Xi\|_l = O_P(k_1(p)^{3/2} \sqrt{g(p, d, n)/n})$ and by Lemma 9 we obtain $\|\widehat{\Xi}^* - \Xi\|_l = O_P(k_1(p)^{3/2} \sqrt{g(p, d, n)/n})$, where $\widehat{\Xi}^*$ is the thresholded estimator based on $\widehat{\Xi}^{*(re)}$ as in (7). Similarly, we obtain $\|\widehat{\Sigma}_\varepsilon^* - \Sigma_\varepsilon\|_\infty = O_P(k_2(p) \sqrt{g(p, d, n)/n})$ by Assumption 1(iv) and Lemma 10. This implies that the estimator $\widehat{\Gamma}^{*(st)}(0)$ which is similarly constructed as $\widehat{\Gamma}^{(st)}(0)$ in (6) but based on $(\widehat{A}^{*(s,re)}, s = 1, \dots, d)$ and $\widehat{\Sigma}_\varepsilon^*$ fulfills by Lemma 8 the rate $\|[\widehat{\Gamma}^{*(st)}(0)]^{-1} - [\Gamma^{(st)}(0)]^{-1}\|_\infty = O_P(k_2(p) k_1(p)^{4.5} k_3(p)^2 \sqrt{g(p, d, n)/n})$. This implies $\Sigma_{\varepsilon;jj} e_{(s-1)p+r}^\top \Gamma^{*(st)}(0) \times e_{(s-1)p+r} = \Sigma_{\varepsilon;jj} e_{(s-1)p+r}^\top \Gamma^{(st)}(0) e_{(s-1)p+r} + o_P(1)$. Since the Gaussian i.i.d. pseudo innovations ε_t^* fulfill the moment condition of Assumption 1(vi) and the bootstrap analogue of the estimators fulfill the required rates, we obtain by the same arguments as those used in the proof of Theorem 1 that, as $n \rightarrow \infty$, $\sqrt{n}(\widehat{A}_{j;r}^{*(s,de)} - \widehat{A}_{j;r}^{(s)}) \xrightarrow{D} \mathcal{N}(0, s.e.(j, r, s)^2)$ in probability. $\square$

Proof of Corollary 4. Follows by Lemma 2 and the arguments used in the proof of Theorem 3 which gives $\|\Sigma_\varepsilon^* - \Sigma_\varepsilon\|_\infty = o_P(1)$ and $\|\widehat{\Gamma}^{*(st)}(0) - \Gamma^{(st)}(0)\|_\infty = o_P(1)$. $\square$

Proof of Theorem 6. By the Cramér–Wold device and the same arguments as those used in the proof of Theorem 1, it follows that the vector $(\sqrt{n}|\widehat{A}_{j;r}^{*(s,de)}|/\widehat{s.e.}(j, r, s) : (j, r, s) \in G)^\top$ converges weakly to a $|G|$ -dimensional normal distribution with zero mean vector and covariance matrix having components those given in Lemma 2. Then, following the same arguments as those used in the proof of Theorem 3 and by Lemma 5, the same result can be established for the limiting distribution of the bootstrap analogue $(\sqrt{n}|\widehat{A}_{j;r}^{*(s,de)}|/\widehat{s.e.}^*(j, r, s) : (j, r, s) \in G)^\top$. The assertion of the theorem follows then by the continuous mapping theorem. $\square$

Proof of Theorem 7. Let $n_g = |G_n|$ and $\mathcal{I} = \{1, \dots, n_g\}$ be a set of indices corresponding to G_n . We show this by using first the triangular inequality, thus

$$\begin{aligned} & \sup_{c \in \mathbb{R}} \left| P^* \left(\max_{(j,r,s) \in G_n} \left| \frac{\sqrt{n}|\widehat{A}_{j;r}^{*(s,de)}|}{\widehat{s.e.}^*(j, r, s)} \right| \leq c \right) - P \left(\max_{(j,r,s) \in G_n} \left| \frac{\sqrt{n}|\widehat{A}_{j;r}^{(s,de)}|}{\widehat{s.e.}(j, r, s)} \right| \leq c \right) \right| \\ & \leq \sup_{c \in \mathbb{R}} \left| P \left(\max_{(j,r,s) \in G_n} \sqrt{n}|\widehat{A}_{j;r}^{(s,de)}|/\widehat{s.e.}(j, r, s) \right) - P \left(\max_{i \in \mathcal{I}} e_i^\top D_0^{-1} Z e_i \geq c \right) \right| \\ & \quad + \sup_{c \in \mathbb{R}} \left| P^* \left(\max_{(j,r,s) \in G_n} \left| \frac{\sqrt{n}|\widehat{A}_{j;r}^{*(s,de)}|}{\widehat{s.e.}^*(j, r, s)} \right| \leq c \right) - P \left(\max_{i \in \mathcal{I}} e_i^\top D_0^{-1} Z e_i \geq c \right) \right|, \end{aligned}$$

where Z is some appropriate Gaussian process and $D_0 \in \mathbb{R}^{n_g \times n_g}$ specified later on. If we can show that the first term is $o(1)$ and the second $o_P(1)$ the assertion follows. We begin with the first term. Note that Lemma 14 gives us $\max_{(j,r,s) \in G_n} (\sqrt{n}|\widehat{A}_{j;r}^{(s,de)}| - A_{j;r}^{(s)}|\widehat{s.e.}(j, r, s)|) = \max_{(j,r,s) \in G_n} |(1/\sqrt{n} \sum_{t=d}^{n-1} Y_{t;(r,j,s)})| + o_P(1)$, where $Y_{t;(r,j,s)} = e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1;j}/s.e.(r, j, s)$. Let $D_0 = \text{diag}(s.e.(r_i, j_i, s_i), i \in \mathcal{I})$. We have $\text{Var}(1/\sqrt{n} \sum_{t=1}^n D_0(Y_{t;(r_i, j_i, s_i)}), i \in \mathcal{I})^\top = \Sigma_T = (\sigma_{T;i_1, i_2}, i_1, i_2 \in \mathcal{I})$, where $\sigma_{T;i_1, i_2} = \Sigma_{\varepsilon, j_{i_1} j_{i_2}} e_{(s_{i_1}-1)p+r_{i_1}}^\top (\Gamma^{(st)}(0))^{-1} e_{(s_{i_2}-1)p+r_{i_2}}$. To see this, note that $\{Y_t\}$ is an uncorrelated sequence and we have $1/n \text{Cov}(s.e.(r_1, j_1, s_1) \sum_{t=1}^n Y_{t;(r_1, j_1, s_1)}, s.e.(r_2, j_2, s_2) \sum_{t=1}^n Y_{t;(r_2, j_2, s_2)}) = 1/n \sum_{t=1}^n \text{Cov}(e_{(s_1-1)p+r_1} (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1;j_1}, e_{(s_2-1)p+r_2} (\Gamma^{(st)}(0))^{-1} W_t \varepsilon_{t+1;j_2}) = \sigma_{T;i_1, i_2}$.

Thus, for the process $Y := \{(Y_{t;(r,j,s)})_{(r,j,s) \in G_n}, t \in \mathbb{Z}\}$ we obtain by Lemma 11 that we have for some $\alpha > 1/2$ $\Psi_{q,\alpha} = O(1)$, $\Upsilon_{q,\alpha} = O(\min(n_g^{1/q}, k_1(p)k_3(p) \log(n_g)^{3/2}))$, and $\|Y\|_{q,\alpha} = O(1)$. For q large enough we have $O(\min(n_g^{1/q}, k_1(p)k_3(p) \log(n_g)^{3/2})) = O(n_g^{1/q})$. Thus, Remark 2 in Zhang and Wu [45] can be applied which gives by Theorem 3.2 in Zhang and Wu [45] an Gaussian approximation if $n_g n^{1-q/2} (\log(n_g n))^{3q/2} = o(1)$. This holds by Assumption 1(vi) and the limit on b . Applying now Theorem 3.2 in Zhang and Wu [45] and using that the other terms are asymptotically negligible, we obtain $\sup_{c \in \mathbb{R}} |P(\max_{(j,r,s) \in G_n} \sqrt{n}|\widehat{A}_{j;r}^{(s,de)}|/\widehat{s.e.}(j, r, s)) - P(|\max_{i \in \mathcal{I}} e_i^\top D_0^{-1} Z e_i| \geq c)| = o(1)$, where $Z = (Z_i, i \in \mathcal{I}) \sim \mathcal{N}(0, \Sigma_T)$. In the following we show that $\sup_{c \in \mathbb{R}} |P^*(\max_{(j,r,s) \in G} |(\sqrt{n}|\widehat{A}_{j;r}^{*(s,de)}|)/(\widehat{s.e.}^*(j, r, s))| \leq c) - P(|\max_{i \in \mathcal{I}} e_i^\top D_0^{-1} Z e_i| \geq c)| = o_P(1)$. Following the arguments used in the proof of Theorem 3, we obtain $\|\Xi^{*(re)} - \Xi\|_l = O_P(k_1(p)^{3/2} \sqrt{g(p, d, n)/n})$, $\|\Sigma_\varepsilon^* - \Sigma_\varepsilon\|_\infty = O_P(k_2(p) \sqrt{g(p, d, n)/n})$ and $\|(\widehat{\Gamma}^{*(st)}(0))^{-1} - (\Gamma^{(st)}(0))^{-1}\|_\infty = O_P(k_2(p)k_1(p)^{4.5}k_3(p)^2 \sqrt{g(p, d, n)/n})$, where the bootstrap analogue of the estimators is denoted by $*$. Hence, Lemma 14 implies $\max_{(j,r,s) \in G_n} (\sqrt{n}|\widehat{A}_{j;r}^{*(s,de)}|)/(\widehat{s.e.}^*(j, r, s)) = \max_{(j,r,s) \in G_n} 1/\sqrt{n} \sum_{t=d}^{n-1} Y_{t;(r,j,s)}^* + o_P(1)$, where $Y_{t;(r,j,s)}^* = e_{(s-1)p+r}^\top (\Gamma^{(st)}(0))^{-1} W_t^* \varepsilon_{t+1;j}^*/$

s.e.(r, j, s). Furthermore, we have

$$\begin{aligned} & \frac{1}{n} \text{Cov}^* \left(\sum_{t=d}^{n-1} e_{(s_1-1)p+r_1}^\top (\Gamma^{(st)}(0))^{-1} W_t^* \varepsilon_{t+1;j_1}^*, \sum_{t=d}^{n-1} e_{(s_2-1)p+r_2}^\top (\Gamma^{(st)}(0))^{-1} W_t^* \varepsilon_{t+1;j_2}^* \right) \\ &= (n-d)/n e_{(s_1-1)p+r_1}^\top (\Gamma^{(st)}(0))^{-1} \widehat{\Gamma}^{(st)}(0) (\Gamma^{(st)}(0))^{-1} e_{(s_2-1)p+r_2} \widehat{\Sigma}_{\varepsilon;j_1 j_2} \\ &= e_{(s_1-1)p+r_1}^\top (\Gamma^{(st)}(0))^{-1} e_{(s_2-1)p+r_2} \Sigma_{\varepsilon;j_1 j_2} + o_P(1) = \sigma_{T;i_1, i_2} + o_P(1), \end{aligned}$$

if $(j_1, r_1, s_1) = (j_{i_1}, r_{i_1}, s_{i_1})$ and $(j_2, r_2, s_2) = (j_{i_2}, r_{i_2}, s_{i_2})$. Thus, $\{Y_{t;(r,j,s)}^*, (r, j, s) \in G_n, t \in \mathbb{Z}\}$ possesses asymptotically the same autocovariance structure as $\{Y_{t;(r,j,s)}, (r, j, s) \in G_n, t \in \mathbb{Z}\}$. This implies that Theorem 3.2 in Zhang and Wu [45] can be applied in the same way as above to $\max_{(j,r,s) \in G_n} 1/\sqrt{n} \sum_{t=d}^{n-1} Y_{t;(r,j,s)}^*$. We obtain $\sup_{c \in \mathbb{R}} |P^*(\max_{(j,r,s) \in G_n} |(\sqrt{n} \widehat{A}_{j;r}^{*(s, \text{de})})|/\widehat{s.e.}^*(j, r, s)| \leq c) - P(|\max_{i \in \mathcal{I}} D_0^{-1} Z e_i| \geq c)| = o_P(1)$. $\square$

Lemma 15. Let $X_t = \sum_{j=0}^{\infty} B_j \varepsilon_{t-j}$, $t \in \mathbb{Z}$, where the ε_t 's are i.i.d., $E \varepsilon_1 \varepsilon_1^\top = \Sigma$, $\sum_{j=0}^{\infty} \|B_j\|_2 \leq C_1 < \infty$ and $(E(v^\top \varepsilon_0)^q)^{1/q} \leq C_2 < \infty$ for all $\|v\|_2 < \infty$, $C_1, C_2 < \infty$. Then, $(E\|v^\top X_1\|^q)^{1/q} < \infty$ for $\|v\|_2 < \infty$.

Lemma 16. Let $C \in \mathbb{R}^{p \times p}$ be some positive definite matrix and $\beta_r^\dagger = e_r - I_{p;-r} \beta_{r;-r}$, where $\beta_{r;-r} = (I_{p;-r}^\top C I_{p;-r})^{-1} (I_{p;-r}^\top C e_r)$. Then, $\beta_r^\dagger = (e_r^\top C^{-1} e_r)^{-1} C^{-1} e_r$.

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Supplementary Material

Supplement A–F (DOI: [10.3150/20-BEJ1239SUPP](https://doi.org/10.3150/20-BEJ1239SUPP); .pdf). We provide additional details to stacked VAR(d) processes, the bias-corrected bootstrap percentage points, the data description, the simulation set-up, additional proofs and a comparison of different sparsity settings.

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